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Influence of the Lattice Boltzmann Collision Operator on the Stability and Accuracy of Particulate Flow Simulations

Jonas Plewinski

Master's Thesis

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Abstract

In this thesis, the influence of the lattice Boltzmann collision operator on numerical stability and accuracy of particulate flow simulations is analyzed. It is well-known that the collision operator of the lattice Boltzmann method (LBM) has a vital role regarding numerical stability for pure fluid flow simulations. The research on the significance of this crucial part of the LBM in reference to a coupled algorithm using the LBM as a fluid flow solver is scarce. On these grounds, a four-way coupled algorithm to simulate particulate flows is used and first analyzed concerning the current state of numerical stability when used for problems with high Reynolds numbers.

Algorithmic changes, such as changing the collision operator of the LBM in the coupled algorithm, have to be accompanied by validation processes to ensure ongoing correctness. Therefore, different collision operators are tested on their influence on the collision behavior between a particle and a wall and validated with measurement data from a similar laboratory experiment.

After the validation process, the collision operators are analyzed regarding their numerical stability with a simulation setup with high Stokes and Reynolds numbers.

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Chapter 1

Introduction

In today's age, the rapid development of the hardware for clusters and supercomputers, thus increasing the performance of such systems, enables unprecedented possibilities to simulate particulate flows with numerous particles. This is especially helpful to not only understand complex phenomena, like sediment erosion [1, 2], the influence of particles in a turbulent fluid environment [3, 4], the development and motion of dunes [5, 6], or fluidization processes [7, 8], but also to further investigate and predict such systems. Direct Numerical Simulations (DNS) enable the analysis of detailed information of forces, trajectories, etc. of every singular particle involved in a simulation. Consequently, engineers have access to information that is impossible to produce in laboratory experiments.

In order to produce this information, a suitable algorithm has to be used. Most data for such systems can be gained by using a four-way coupled algorithm, i.e., one that considers every possible interaction that can occur, fluid-particle, particle-fluid, and particle-particle interactions.

Mostly, a direct discretization of the Navier-Stokes equations as a fluid flow solver which is coupled with an immersed boundary method (IBM) to resolve fluid-particle and particle-fluid interactions. Often, a soft contact model collision model accounts for particle-particle interactions [1–4]. These *classical approaches* are well conducted, calibrated, and validated, which is necessary especially for simulations where numerous particle collisions are involved [9–11] to resolve collision behavior as close as possible to the real world. Such calibrations and validations are laborious and time-consuming.

The lattice Boltzmann method (LBM) is an alternative to the classical DNS approaches for the simulation of fluid flow. Even though LBM does not solve the NSE directly but the discrete Boltzmann equation, it equally captures the underlying physics. In addition, it is well suited for highly parallel simulations. Apart from a modified version of the IBM to be usable in combination with the LBM, there exist other LBM specific methods, such as the discrete element method (DEM) modified for LBMs to also cover particle-particle interactions. This enables studies on complex particle-laden flows in large-scale simulations [8, 12, 13].

As mentioned before, rigorous testing of the used methods to simulate particulate flows is indispensable. This was already done for classical DNS approaches but is almost non-existent for the LBM. Such studies exist for pure fluid flows or multiphase flows with the LBM [14–18]. However, apart from the calibration and validation pipeline developed by Rettinger in [19] for assessing the collision behavior of coupled algorithms used for simulating particulate flows, we are unaware of such studies. Especially studies regarding the influence of the LBM-specific collision operator on the accuracy of the collision behavior and numerical stability does not exist. The collision operator plays a vital role in the LBM and influences accuracy and numerical stability for pure fluid flows and multiphase flows. Still, it is unknown if the collision operator has the same impact on particulate fluid simulations.

Often only one collision operator is used for a specific application and then optimized. Rarely multiple collision operators are directly compared against one another, because the efficient implementation of advanced collision operators like the multiple-relaxation time or cumulant collision operators can be time-consuming and complex. We can avoid this problem by using the code generation tool *lbmpy* to generate efficient LBM kernels in only a few lines of python code. Together with the high-performance computing C++ framework waLBerla, collision operators are easily interchangeable to enable the possibility to test, validate and compare different collision operators against one another.

The thesis is structured in the following way: In Chapter 2, the four-way coupled algorithm developed by Rettinger in [19] is presented, consisting of the LBM for simulation of the fluid phase, the moment exchange method (MEM) enabling the coupling between fluid and particle, a lubrication model capturing the processes when particles come close to one another, but are still not in contact and the discrete element method (DEM) resolving the direct particle-particle or particle-wall interactions. Afterwards in Chapter 3, the coupled algorithm is analyzed regarding its numerical stability. Subsequently, different collision operators in the coupled algorithm are investigated concerning their collision behavior with a well-studied sphere-wall collision test case and compared against experimental measurement data in Chapter 4. Lastly the same collision operators are analyzed concerning their numerical stability with the same simulation setup but increased Stokes numbers, in Chapter 5.

Chapter 2

Numerical Methods

In Chapter 2, the specific parts of the four-way coupled algorithm to simulate particulate flows are introduced. First the lattice Boltzmann method used for solving and simulating the fluid flow part presented in Section 2.1. Necessary parts like the streaming and collision step are described as well as different collision operators representing particle distribution interactions in the collision step. Coupling the fluid flow simulation with the explicitly mapped particles is done via the Momentum-Exchange method described in Section 2.2. Before particles collide with one another or colliding with a wall the present fluid between the participants of such a collision is pressed out, possibly resulting resistances, which have to be modeled to simulate a correct collision behavior. This is introduced in Section 2.3. The discrete element method is handling the actual particle-particle or particle-wall collisions, which is defined in Section 2.4 before summarizing all parts of the four-way coupled algorithm in a pseudo code shown in Section 2.5.

2.1 The Lattice Boltzmann Method

There are various ways to simulate fluid flow in computational fluid dynamics (CFD). Often a model following a macroscopic view is used for such simulations. In these cases, the fluid is represented as a continuum and not as, e.g., particles or molecules as it would be the case in a microscopic view. The Navier Stokes equations (NSE) follow such a macroscopic view. The NSE for an incompressible fluid consist of a scalar continuity and a vectorial momentum equation defined as

$$\nabla \cdot \boldsymbol{u} = 0 \tag{2.1a}$$

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla p + \nabla \cdot \left(\mu \left[\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^{\mathrm{T}} \right] \right) + \boldsymbol{F}. \tag{2.1b}$$

Here, $\boldsymbol{u}$ is given as the fluid velocity, ρ describes the density of the fluid, p the pressure, μ the dynamic viscosity, and the external forces represented by $\boldsymbol{F}$, e.g., gravity. Unfortunately, there exists no general analytical solution for these equations

until today. Therefore the NSE have to be approximated via numerical schemes, like the finite difference or the finite volume method [20].

Another approach to describe the development of fluid flow uses the Boltzmann equation. It follows a mesoscopic view of the fluid, originating from statistical mechanics, and describes the evolution of particle distribution functions (PDFs) on a regular grid. Here, instead of tracking every particle of such a PDF separately, as it is done in the microscopic view, their statistical properties are investigated. The PDF $f(\boldsymbol{x}, \boldsymbol{\xi}, t)$ describes the density of particles at position $\boldsymbol{x}$ with a microscopic velocity $\boldsymbol{\xi}$ at the time t [20]. By calculating moments of the PDFs, macroscopic quantities, like the mass density and momentum density can be recovered via

$$\rho(\boldsymbol{x}, t) = \int f(\boldsymbol{x}, \boldsymbol{\xi}, t)\, \mathrm{d}^3\xi$$
$$\rho(\boldsymbol{x}, t)\, \boldsymbol{u}(\boldsymbol{x}, t) = \int \boldsymbol{\xi}\, f(\boldsymbol{x}, \boldsymbol{\xi}, t)\, \mathrm{d}^3\xi$$

The fluid's evolution in time can be described by the temporal derivative of its fundamental variable, the PDF, yielding the continuous Boltzmann equation

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \xi_\beta \frac{\partial f}{\partial x_\beta} + \frac{F_\beta}{\rho} \frac{\partial f}{\partial \xi_\beta} = \Omega(f). \tag{2.2}$$

The advection of the PDF f with the microscopic velocity $\boldsymbol{\xi}$ is represented by the first two terms. The third term contains external forces, and the right-hand side $\Omega(f)$ describes the source term, which models the local redistribution of the PDF f due to collisions. Hence, this source term is also called *collision operator*. It must respect mass, momentum, total, and internal energy conservation. In addition, assuming that after a sufficient number of collisions, the particles are distributed isotropically around the mean velocity $\boldsymbol{u}$ [20], it needs to ensure that the local PDF evolves locally to its equilibrium f^{eq} [20].

In order to solve the lattice Boltzmann equation numerically, a discretization in physical space, velocity space, and time is necessary, which results in the lattice Boltzmann equation (LBE)

$$f_q(\boldsymbol{x} + \boldsymbol{c}_q \Delta t, t + \Delta t) = f_q(\boldsymbol{x}, t) + \Omega_q(\boldsymbol{x}, t). \tag{2.3}$$

This formula describes the movement of PDFs $f_q(\boldsymbol{x}, t)$ with the discrete velocity $\boldsymbol{c}_q$ to a neighboring point $\boldsymbol{x} + \boldsymbol{c}_q \Delta t$ at the following time step $t + \Delta t$. Simultaneously, particles are influenced by collisions. These are modeled by redistributing particles between populations at each site, which is represented by the collision operator $\Omega_q(\boldsymbol{x}, t)$ [20]. This collision operator will be further discussed in Section 2.1.1.

The computation of the LBE is divided into two distinct parts, which are performed in succession for every time step:

1. **Collision Step / Relaxation**
   The first part is defined as the collision step, also called relaxation:
   $$f_q^*(\boldsymbol{x}, t) = f_q(\boldsymbol{x}, t) + \Omega_q(\boldsymbol{x}, t). \tag{2.4}$$
   Here, f_q^* denotes the discrete distribution function after the collision step. This operation is entirely local [20, p, 66].
2. **Streaming Step / Propagation**
   The second part is called the streaming step or propagation:
   $$f_q(\boldsymbol{x} + \boldsymbol{c}_q \Delta t, t + \Delta t) = f_q^*(\boldsymbol{x}, t). \tag{2.5}$$
   Here the post-collision distributions f_q^* are streamed to neighboring nodes [20].

Space and time resolution are described by the lattice spacing Δx and time step Δt, respectively. Any set of units can be used for these. However, most commonly, so-called lattice units are used. These are artificial units, where $\Delta x = 1$ and $\Delta t = 1$. We exploit the law of similarity to ensure that two simulations with different units are simulating the same behavior. Using this law, we have to confirm that relevant dimensionless characteristic numbers[1] are identical [20].

The stencil given as DdQq represents a specific velocity set. Here d defines the number of dimensions and q the number of discrete velocities. Typical stencils used for the simulation of 3-dimensional problems are the D3Q15, D3Q19, and the D3Q27 stencil. The decision on which stencil is used depends on the fluid flow problem at hand. While smaller stencils are computationally cheaper, they rather suffer from instabilities due to numerical errors than larger stencils. In particular, looking at turbulent simulations with high Reynolds numbers, instabilities are hard to overcome. Hence the best choice for simulating such problems is the D3Q27 stencil [20].

Analogously to the continuous representation, the macroscopic variables are recovered as moments of f_q, i.e. [20]

$$\rho(\boldsymbol{x}, t) = \sum_q f_q(\boldsymbol{x}, t), \tag{2.6a}$$

$$\rho(\boldsymbol{x}, t)\, \boldsymbol{u}(\boldsymbol{x}, t) = \sum_q \boldsymbol{c}_q f_q(\boldsymbol{x}, t). \tag{2.6b}$$

2.1.1 Multiple–Relaxation–Time Collision Operator

The choice of the collision operator of the LBM is essential regarding the stability and accuracy of the fluid flow simulation. The core of the collision operator is the relaxation of the PDFs towards their equilibrium. Generally, there are three possibilities where the collision is taking place, in the

[1] E.g., Reynolds number, Galileo number or Stokes number. These are characteristic dimensionless numbers for different kinds of fluid flow problems.

- velocity space containing the PDFs
- moment space or
- cumulant space, see Section 2.1.4.

Most of the collision operators transform the PDFs into moments to perform the collision in the moment space [21], where it can be written as

$$\Omega = \boldsymbol{M}^{-1}\boldsymbol{S}\left(\boldsymbol{m}^{\mathrm{eq}} - \boldsymbol{M}\boldsymbol{f}\right) \tag{2.7}$$

Here, $\boldsymbol{M}$ is called the moment transformation matrix, $\boldsymbol{S}$ contains the relaxation rates and the vector $\boldsymbol{m}^{\mathrm{eq}}$ holds the equilibrium moments. This approach is called the multiple-relaxation-time (MRT) model because it enables the relaxation of different moments with different relaxation rates.

This work is mostly based on Rettinger *et al.* There, the D3Q19–MRT model of Schiller and Ladd [22] was used, where the diagonal relaxation rate matrix $\boldsymbol{S}$ can be defined as

$$\boldsymbol{S} = \mathrm{diag}(0,\, 0,\, 0,\, 0,\, s_1,\, s_2,\, s_3,\, s_3,\, s_3,\, s_4,\, s_4,\, s_4,\, s_4,\, s_4,\, s_5,\, s_5,\, s_6,\, s_6,\, s_6). \tag{2.8}$$

In order to conserve moments for mass and momentum, the first four entries of $\boldsymbol{S}$ have to be set to zero, such that the first four moments do not change during the collision.

The fluid kinematic viscosity v_f is calculated via the relaxation rate of the shear moments, given by s_v:

$$v_f = \frac{1}{3}\left(\frac{1}{s_v} - \frac{1}{2}\right). \tag{2.9}$$

Determining the bulk viscosity v_b is done analogously:

$$v_b = \frac{2}{9}\left(\frac{1}{s_b} - \frac{1}{2}\right). \tag{2.10}$$

with s_b being the relaxation rate of the bulk moment [19]. Using the MRT model from [19] and [22] the relaxation rates s_4 and s_1 from Equation (2.8) are set to s_v and s_b respectively. The other relaxation rates contained in $\boldsymbol{S}$ can be used to optimize the MRT model regarding numerical accuracy or stability without affecting the underlying physics [19]. Finding the optimal choice for a given problem is time-consuming due to the calibration procedure [20].

2.1.2 Multiple–Relaxation–Time Specializations

With the general definition of the collision operator in Equation (2.7), it is possible to formulate common variants of the LBM as special cases. By using the same relaxation rates for all moments, we can recover the so-called single-relaxation-time (SRT) or Bhatnagar-Gross-Krook (BGK) model [23], which leads to

$$\boldsymbol{S}^{\mathrm{SRT}} = \mathrm{diag}(0,\, 0,\, 0,\, 0,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v). \tag{2.11}$$

Although the SRT model is the most popular due to its simplicity, it has well-known drawbacks concerning accuracy and stability [20]. Specifically, the unwanted dependency of the boundary location on the fluid viscosity due to the relaxation rate problematic and was addressed in [24].

A slightly more complex collision operator, the two-relaxation-time (TRT) model aims to obtain viscosity-independent boundary locations. To get these, we first introduce a second relaxation rate s_v^-, which is used for all odd-order moments in Equation (2.11), such that we gain

$$\boldsymbol{S}^{\mathrm{TRT}} = \mathrm{diag}(0,\, 0,\, 0,\, 0,\, s_v,\, s_v,\, s_v^-,\, s_v^-,\, s_v^-,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v^-,\, s_v^-,\, s_v^-).$$

Additionally, the following relation has to be satisfied by both relaxation rates

$$\Lambda = \left(\frac{1}{s_v} - \frac{1}{2}\right)\left(\frac{1}{s_v^-} - \frac{1}{2}\right). \tag{2.12}$$

Λ defines the "magic"-parameter, a constant, which characterizes the stability properties and the truncation error of the TRT model [20]. Here we set it to $\Lambda = 3/16$.

2.1.3 Advanced Collision Operator for an Independent Control of the Bulk Viscosity

We employ the MRT model developed by Ladd et. al. in [22]. The transformation matrix used in Equation (2.7), as well as its equilibrium values, are stated in Appendix A.1.

This MRT model was further developed by Rettinger in [19]. It is an extended D3Q19 TRT model, called the TRT+B model, which can control the bulk viscosity, independently from the shear viscosity. This can be done without forfeiting the advantages gained by using the TRT model. Following [25], we introduce an additional relaxation rate s_b for the kinetic energy and kinetic energy square moment, i.e.,

$$\boldsymbol{S}^{\mathrm{TRT+B}} = \mathrm{diag}(0,\, 0,\, 0,\, 0,\, s_b,\, s_b,\, \bar{s_v},\, \bar{s_v},\, \bar{s_v},\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, s_v,\, \bar{s_v},\, \bar{s_v},\, \bar{s_v}).$$

For stability reasons not only s_1 is set to s_b, but also s_2. The relation

$$\Lambda_b = \frac{\frac{1}{s_b} - \frac{1}{2}}{\frac{1}{s_v} - \frac{1}{2}}$$

is used to guarantee viscosity-independent boundary locations [19]. Here, Λ_b impacts the behavior of the bulk viscosity:

- $\Lambda_b = 1$: standard TRT model
- $\Lambda_b > 1$: bulk viscosity increases
- $\Lambda_b < 1$: bulk viscosity decreases

Rettinger showed in [19] that choosing $\Lambda_b = 10$ damps nonphysical oscillations effectively to gain better stability with this TRT+B model. This model was further developed by introducing the increased bulk viscosity only for a shell with a given thickness of $\delta_{\mathrm{aB}} = 2\Delta x$ around all particles called TRT+aB. With this approach, the sphere-specific information is as local as possible and at any given moment, the layer size of the shell is at least one cell thick [19]. Accordingly, Λ_b is set to 100 inside the shell and 1 everywhere else.

2.1.4 Cumulant Collision Operator

Due to the complexity of the cumulant collision operator, we will only present the main idea here. Further information can be gained by reading [14, 26, 27].

In the case of the SRT collision operator, the collision takes place in the velocity space of the distribution functions. For the MRT, the PDFs must be transformed to the moment space before the collision. Afterward, the post-collision PDFs have to be transformed back to the space of distribution functions for the streaming step. Specific physical quantities like density, momentum, or energy can be assigned to specific moments with this approach. From a physical point of view, the MRT collision operator is superior to the SRT because it considers several distribution functions and their possible interactions at once for each moment.

On the one hand, one tries to consider as many interactions with other PDFs or moments as possible. On the other hand, one tries to separate the moments from each other in a certain way, e.g., by orthogonalizing them. This approach is justified because the associated physical quantities strive towards their equilibrium according to different laws and speed. The fact that these quantities are influenced by possibly even independent processes should ideally be considered, and individual evolution should be enabled. Let us shortly illustrate this with an easy example: The energy E of particles can be assigned to one moment, while the energy square E^2 can be assigned to another moment. Hypothetically, if the energy now has a value of 5, the probability that the energy square is 25 is 100%. These two moments are stochastically dependent on each other. Therefore, the question arises, if it is possible to generate new moments by a transformation of any kind, which results in distribution functions containing statistically independent quantities. One example for such statistically independent quantities are the so-called cumulants.

It is possible to transform PDFs directly into cumulants. Unfortunately due to cumulants being no simple functions, this approach is accompanied by an extreme computational cost. Hence in practice, a multi-step process is used. The PDFs are transformed into the central moment space, before transforming the resulting central moments into the cumulant space. Central moments use an intrinsic frame of reference to gain independence of a given system's frame of reference and thus achieve a better

Galilean invariance[2]. This independence is obtained by shifting raw moments by the macroscopic fluid velocity. Discrete central moments are given by

$$K_{\alpha\beta\gamma} = \sum_q \Big((c_q)_x - u_x \Big)^\alpha \Big((c_q)_y - u_y \Big)^\beta \Big((c_q)_z - u_z \Big)^\gamma f_q(\boldsymbol{x}, t) \,.$$

Here, the summation of $\alpha + \beta + \gamma$ represent the total order of the moment $K_{\alpha\beta\gamma}$, $(c_q)_x$ describes the velocity components of c_q in the x-direction and u_x the macroscopic fluid velocity in the x-direction.

Now we will give a rough overview of the process using cumulants:

1. Transform PDFs into centered moments via forward central moment transformation.
2. Shift the centered moments via non-linear forward cumulant transformation in the cumulant space to compute the cumulants.
3. If specific cumulants do not respect rotational invariance, form linear combinations of cumulants and relax them while respecting rotational invariance. Use the terms to achieve Galilean invariance. If the linear combinations are chosen appropriately, the equilibrium values become zero.
4. In theory, a linear system of equations has now to be solved in every time step to obtain the individual cumulants from the linear combinations after the relaxation. But as a consequence of choosing the linear combinations to get zero-equilibrium values, this is not necessary in practice.
5. Transform post-collision cumulants back to relaxed centered moments via backward cumulant transformation.
6. Transform relaxed centered moments back to post-collision distribution functions via backward central moment transformation.
7. Perform the streaming step.

As the transformation into the cumulant space is non-linear we cannot use a matrix-vector notation as it is typical for the moment base collision operators. For the D3Q27 cumulant stencil, ten relaxation rates are used ω_x for $x \in [0, \ldots, 9]$, where only ω_0 is depending on the kinematic viscosity. The other nine relaxation rates have an influence on higher-order moments. This influence is varying depending on their value and can be freely chosen in the range of [0,. . . ,2]. Setting them to one is sufficient for most applications.

[2]Galilean invariance refers to a state where in any inertial reference frame the physical laws are equal.

2.2 Fluid-Particle Coupling via the Momentum-Exchange Method

For the two-way coupling of the fluid flow and the particles, the LBM-specific momentum exchange method (MEM) is used. This method was defined by Ladd [28] and further developed by Aidun *et al.* [29]. This method introduces boundary conditions along the particle's surface for the fluid and calculates a hydrodynamic interaction force $\boldsymbol{F}_{p,i}^{fp}$ and torque $\boldsymbol{T}_{p,i}^{fp}$, which act on particle i.

In this approach, all particles are mapped explicitly into the fluid domain. As a result, all cells with a cell center outside a particle are considered fluid cells. These cells make up the fluid flow simulation domain, where the LBM is solved. Whereas cells with a cell center inside a particle are considered solid cells. They do not carry fluid information [29] as is shown in the left part of Figure 2.1.

We use the central line interpolation (CLI), see Figure 2.1, to establish the interaction between particles and fluid. This is a higher-order no-slip boundary condition, which uses subgrid information to represent the explicitly mapped particles' surface with a higher resolution. In doing so, we obtain a better coupling accuracy [30]. It is defined by

$$f_{\bar{q}}\left(\boldsymbol{x}, t+\Delta t\right) = \tilde{f}_q\left(\boldsymbol{x}, t\right) + \frac{1-2\delta_q}{1+2\delta_q}\left(\tilde{f}_q\left(\boldsymbol{x}-\boldsymbol{c}_q\Delta t, t\right) - \tilde{f}_{\bar{q}}\left(\boldsymbol{x}, t\right)\right) - \frac{4}{1+2\delta_q}\, 3 w_q \rho_0 \boldsymbol{u}_b \cdot \boldsymbol{c}_q \tag{2.13}$$

with $\bar{q}$ representing the opposite lattice direction of q, i.e., $-\boldsymbol{c}_q = \boldsymbol{c}_{\bar{q}}$. The lattice weights are given by w_q [31], the boundary velocity by $\boldsymbol{u}_b = \boldsymbol{U}_i\left(\boldsymbol{x}_b\right)$ with

$$\boldsymbol{U}_i\left(\boldsymbol{x_b}\right) = \boldsymbol{u}_{p,i} + \left(\boldsymbol{x}_b - \boldsymbol{x}_{p,i}\right) \times \omega_{p,i},$$

where $\omega_{p,i}$ describes the angular velocity of particle i. Furthermore, δ_q defines the normalized distance between the exact position of the surface and the cell center, such that $\boldsymbol{x}_b = \boldsymbol{x} + \delta_q \boldsymbol{c}_q \Delta t$ is satisfied. Figure 2.1 shows the basic approach of this boundary condition. For this boundary treatment, we set $\delta_q = 1/2$ in Equation (2.13) to utilize the bounce-back boundary condition if we have insufficient fluid information.

With the momentum-exchange method, we then evaluate the hydrodynamic force and torque acting on the particle [28]. For a given fluid-solid link[3] q, the local interaction force at a boundary location $\boldsymbol{x}_b$ is given by [32]

$$\boldsymbol{F}_q\left(\boldsymbol{x}_b, t\right) = \left(\boldsymbol{c}_q - \boldsymbol{u}_b\right)\tilde{f}_q\left(\boldsymbol{x}, t\right) - \left(\boldsymbol{c}_{\bar{q}} - \boldsymbol{u}_b\right) f_{\bar{q}}\left(\boldsymbol{x}, t+\Delta t\right).$$

[3] Direction, which was treated with the boundary condition defined in Equation (2.13)

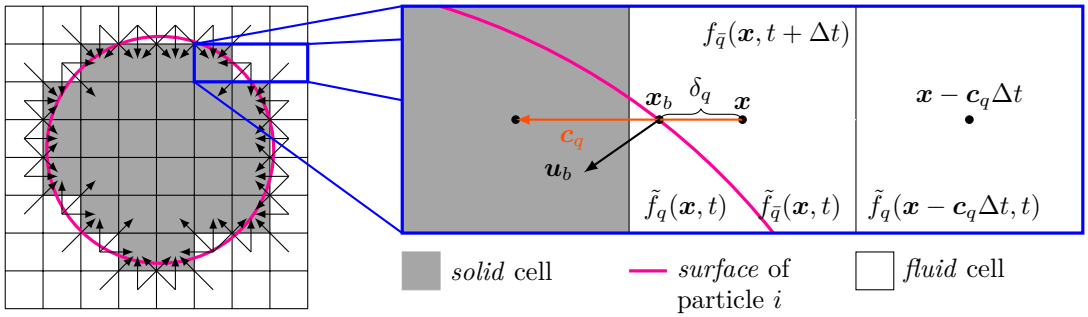


Figure 2.1: *Left part:* Outline of the MEM with explicit particle mapping together with all fluid-solid directions visualized as black arrows. *Right part:* The blue emphasized part shows the CLI boundary condition as stated in Equation (2.13). Sketch on the basis of [19, Figure 2]

The hydrodynamic force and torque acting on particle i can now be calculated as [28]

$$\boldsymbol{F}_{p,i}^{fp}(t) = \sum_{\boldsymbol{x}_b \text{ of } i} \sum_q \boldsymbol{F}_q(\boldsymbol{x}_b, t), \tag{2.14a}$$

$$\boldsymbol{T}_{p,i}^{fp}(t) = \sum_{\boldsymbol{x}_b \text{ of } i} \sum_q (\boldsymbol{x}_b - \boldsymbol{x}_{p,i}) \times \boldsymbol{F}_q(\boldsymbol{x}_b, t). \tag{2.14b}$$

In situations, where two particle surfaces are close to one another, some of these fluid-solid links could be missing [33]. This results in an incomplete momentum-balance of the fluid by introducing an artificial attractive force between the two particles. In these cases we set the local interaction force to the pressure force in stationary conditions, i.e., $\boldsymbol{F}_q(\boldsymbol{x}_b, t) = 2w_q\boldsymbol{c}_q$[4], to account for the missing forces [19].

As a consequence of the explicit particle mapping into the fluid domain, solid cells of moving particles will eventually transition to fluid cells. In this regard, necessary fluid data is missing in the transition cell $\boldsymbol{x}_t$. This information has to be restored before the simulation can proceed [19]. Following the ideas of references [34, 35], we use the local pressure tensor to increase the accuracy and set the missing PDF values in the transition cell $\boldsymbol{x}_t$ with

$$f_q(\boldsymbol{x}_t) = w_q \left(\bar{\rho} + \frac{\bar{\rho}\boldsymbol{u}_{t,\alpha}\boldsymbol{c}_{q\alpha}}{c_s^2} + \frac{1}{2c_s^4} \left(\bar{\rho}\boldsymbol{u}_{t,\alpha}\boldsymbol{u}_{t,\beta} - \frac{\bar{\rho}c_s^2}{s_v} \left(\frac{\partial \boldsymbol{u}_{t,\alpha}}{\partial \boldsymbol{x}_\beta} + \frac{\partial \boldsymbol{u}_{t,\beta}}{\partial \boldsymbol{x}_\alpha} \right) \right) \left(\boldsymbol{c}_{q\alpha}\boldsymbol{c}_{q\beta} - c_s^2\delta_{\alpha\beta} \right) \right) \tag{2.15}$$

where $\delta_{\alpha\beta}$ represents the Kronecker delta, $\bar{\rho}$ a density which is spatially averaged over all neighboring cells, and $\boldsymbol{u}_t$ is initialized with the velocity of the formerly present particle $\boldsymbol{U}_i(\boldsymbol{x}_t)$. If enough data is available, the derivatives of the velocity are approximated via second-order finite differences. Otherwise these are replaced with first-order forward or backward finite differences.

[4] Represents the pressure force in stationary conditions

2.3 Lubrication Model

In situations with two approaching particles or one particle comes close to a wall, the fluid between the surfaces is pressed out. Hence, this can lead to large resistance since that lubrication forces and torques are acting against the relative motion of the particle, which can significantly affect the particle-particle / particle-wall interaction behavior [19]. Consequently, one has to either use a high resolution to accurately capture this behavior or use some kind of lubrication correction model [9, 10, 19, 36, 37]. Because the first option is computationally not feasible [36], the second approach is commonly used. In order to use a lubrication correction model, we modify the total hydrodynamic force $\boldsymbol{F}_{p,i}^{\text{hyd}}$ and torque $\boldsymbol{T}_{p,i}^{\text{hyd}}$, such that

$$\boldsymbol{F}_{p,i}^{\text{hyd}} = \boldsymbol{F}_{p,i}^{fp} + \boldsymbol{F}_{p,i}^{\text{lub-corr}}, \tag{2.16a}$$

$$\boldsymbol{T}_{p,i}^{\text{hyd}} = \boldsymbol{T}_{p,i}^{fp} + \boldsymbol{T}_{p,i}^{\text{lub-corr}}. \tag{2.16b}$$

The correction terms $\boldsymbol{F}_{p,i}^{\text{lub-corr}}$ and $\boldsymbol{T}_{p,i}^{\text{lub-corr}}$ are commonly set to be the leading order terms of the analytically computed lubrication forces for the Stokes flow. Consequently, the contributions are split into normal and tangential parts acting on particle i [38, 39]

$$\boldsymbol{F}_{p,i}^{\text{lub-corr}} = \sum_{j \neq i} \left(\boldsymbol{F}_{ij,n}^{\text{lub-corr}} + \boldsymbol{F}_{ij,tt}^{\text{lub-corr}} + \boldsymbol{F}_{ij,tr}^{\text{lub-corr}} \right),$$

$$\boldsymbol{T}_{p,j}^{\text{lub-corr}} = \sum_{j \neq i} \left(\boldsymbol{T}_{ij,tt}^{\text{lub-corr}} + \boldsymbol{T}_{ij,tr}^{\text{lub-corr}} \right),$$

The lubrication correction force in the normal direction acting on particle i, is given by [36–38]

$$\boldsymbol{F}_{ij,n}^{\text{lub-corr}} = -6\pi\mu_f R_{p,i}^2 \lambda \left(\delta_n^{\text{lub}}, \delta_{n,\text{cut}}^{\text{lub}} \right) \frac{\kappa_r^2}{(1+\kappa_r)^2} \left(\frac{1}{\delta_n^{\text{lub}} - \delta_{n,\text{cut}}^{\text{lub}}} \boldsymbol{u}_{ij,n} \right) \tag{2.18}$$

where κ_r defines the radius ratio $R_{p,j}/R_{p,i}$, assuming particles i and j have a different radius[5]. Furthermore, μ_f describes the dynamic fluid viscosity, $\boldsymbol{u}_{ij,n}$ the normal relative velocity of the particle i and δ_n^{lub} the gap size according to

$$\delta_n^{\text{lub}} = \max \left(\delta_{ij,n}, \delta_{n,\text{min}}^{\text{lub}} \right) \tag{2.19}$$

If the gap size is getting very small, δ_n^{lub} tends to zero, which would lead to a division by zero in Equation (2.18). To avoid this, we use a minimal gap size $\delta_{n,min}^{\text{lub}}$ to effectively saturate the lubrication force [19].

[5] For a particle-wall interaction we assume $\kappa_r \to \infty$ [40, 41].

Intending to switch off the lubrication correction during a collision and when the distance between particles is big enough, we define a step function λ as

$$\lambda\left(\delta_n, \delta_{\text{cut}}^{\text{lub}}\right) = \begin{cases} 1, & 0 < \delta_n < \delta_{\text{cut}}^{\text{lub}}, \\ 0, & \text{otherwise} \end{cases}$$

with $\delta_{\text{cut}}^{\text{lub}}$ defining a cut-off distance[6] [19].

The tangential components of the lubrication correction are not relevant for the simulations considered herein and are, therefore, not stated for brevity but can be found in [19].

2.4 Discrete Element Method

Following [19], the particle-particle or particle-wall collision behavior is realized with the discrete element method (DEM). After the computation of all necessary forces and torques which act on the particles, we use the Velocity-Verlet integrator to update the position and velocity of each particle. In this work, we focus on the collision behavior of spherical particles.

2.4.1 Equations of particle motion

Before we take a look at the collision between particles we have to describe the motion of a single particle, which is defined by

$$m_{p,i}\frac{\mathrm{d}\boldsymbol{u}_{p,i}}{\mathrm{d}t} = \boldsymbol{F}_{p,i} = \boldsymbol{F}_{p,i}^{\text{col}} + \boldsymbol{F}_{p,i}^{\text{hyd}} + \boldsymbol{F}_{p,i}^{\text{ext}}, \tag{2.20a}$$

$$I_{p,i}\frac{\mathrm{d}\omega_{p,i}}{\mathrm{d}t} = \boldsymbol{T}_{p,i} = \boldsymbol{T}_{p,i}^{\text{col}} + \boldsymbol{T}_{p,i}^{\text{hyd}}, \tag{2.20b}$$

where $m_{p,i}$ describes the mass of particle i with a given density $\rho_{p,i}$ and volume $V_{p,i}$, such that $m_{p,i}$ is computed via $m_{p,i} = \rho_{p,i}V_{p,i}$. Furthermore, the moment of inertia $I_{p,i}$ for a sphere with a radius $R_{p,i}$ is defined by $I_{p,i} = {}^2\!/_5 m_{p,i}R_{p,i}^2$ [19]. The change of the translational velocity over time of particle i is given by the total force $\boldsymbol{F}_{p,i}$. The temporal evolution of the angular velocity on the other hand, is described by its total torque $\boldsymbol{T}_{p,i}$. These consist of the already known hydrodynamic interactions $\boldsymbol{F}_{p,i}^{\text{hyd}}$ and $\boldsymbol{T}_{p,i}^{\text{col}}$, calculated with Equations (2.16). The force and torque contributions due to inter-particle collisions $\boldsymbol{F}_{p,i}^{\text{col}}$ and $\boldsymbol{T}_{p,i}^{\text{col}}$, will be derived in this section. Additionally, external forces like gravity and buoyancy, given by $V_{p,i}\left(\rho_{p,i} - \rho_f\right)\boldsymbol{g}$ with $\boldsymbol{g}$ being the gravitational acceleration, might also be present.

[6] For all normal distances δ_n, between the surfaces of particles or the surface of a particle and a wall, larger than the one given by $\delta_{\text{cut}}^{\text{lub}}$, the lubrication correction model is switched off.

A time integration via the Velocity-Verlet scheme of Equations (2.20) results in the position and velocity update formulas for particle i [42]

$$\boldsymbol{x}_{p,i}\left(t+\Delta t_p\right)=\boldsymbol{x}_{p,i}\left(t\right)+\Delta t_p\boldsymbol{u}_{p,i}\left(t\right)+\frac{\Delta t_p^2}{2m_{p,i}}\boldsymbol{F}_{p,i}\left(t\right), \tag{2.21a}$$

$$\boldsymbol{u}_{p,i}\left(t+\Delta t_p\right)=\boldsymbol{u}_{p,i}\left(t\right)+\frac{\Delta t_p}{2m_{p,i}}\Big(\boldsymbol{F}_{p,i}\left(t\right)+\boldsymbol{F}_{p,i}\left(t+\Delta t_p\right)\Big) \tag{2.21b}$$

with Δt_p describing the time step size. To obtain $\boldsymbol{F}_{p,i}\left(t+\Delta t_p\right)$ we use the already updated position of particle i.

2.4.2 Contact model for inter-particle collisions

We use the discrete element method (DEM) for the inter-particle collisions, which are split into two parts: the normal and tangential contributions are considered for every pair of interactions [19]. This is defined by

$$\boldsymbol{F}_{p,i}^{\mathrm{col}}=\sum_{j,j\neq i}\Big(\boldsymbol{F}_{ij,n}^{\mathrm{col}}+\boldsymbol{F}_{ij,t}^{\mathrm{col}}\Big), \tag{2.22a}$$

$$\boldsymbol{T}_{p,i}^{\mathrm{col}}=\sum_{j,j\neq i}\Big(\boldsymbol{x}_{ij}^{cp}+\boldsymbol{x}_{p,i}\Big)\times\boldsymbol{F}_{ij,t}^{\mathrm{col}} \tag{2.22b}$$

A linear spring-dashpot model [43] is used for the normal part of the collision force and is given by

$$\boldsymbol{F}_{ij,n}^{\mathrm{col}}=-k_n\delta_{ij,n}\boldsymbol{n}_{ij}-d_n\boldsymbol{u}_{ij,n}^{cp}.$$

Here, k_n describes the normal stiffness, while d_n is the damping coefficient. Instead of defining these directly, we introduce two new variables:

- T_c: the collision time refers to the period of one collision. This parameter has to be chosen carefully because it has a significant influence on the collision behavior of particles [19]. Typical time scales for the fluid are magnitudes greater than for collisions. In choosing $T_c>1$, the collision is stretched in time, such that proper treatment for both the fluid field and the collision is possible. Otherwise, an extremely small time step size would be required for simulations with collisions. Furthermore, by choosing the collision time to be longer, the fluid field around the particle can adapt better to the sudden change of its velocity. Nevertheless, the collision time also introduces an overlap of the surfaces of its collision participants, which should be kept small [9, 10].
- e_{dry}: coefficient of restitution specifies the ratio of normal velocity after and before the collision [44]:

$$e_{\mathrm{dry}}=-\frac{\left(\boldsymbol{u}_{ij,n}\cdot\boldsymbol{n}_{ij}\right)\big|_{\mathrm{post}}}{\left(\boldsymbol{u}_{ij,n}\cdot\boldsymbol{n}_{ij}\right)\big|_{\mathrm{pre}}}$$

With the condition that there is no overlap at the end of the collision between colliding particles, we can compute the motion of the linear harmonic oscillator analytically, resulting in [44]

$$k_n = \frac{m_{ij,\text{eff}}\left(\pi^2 + \ln\left(e_{\text{dry}}\right)\right)}{T_c^2}$$
$$d_n = -\frac{2m_{ij,\text{eff}}\ln\left(e_{\text{dry}}\right)}{T_c}$$

with $m_{ij,\text{eff}}$ describing the effective mass and is given by

$$m_{ij,\text{eff}} = \begin{cases} \frac{m_{p,i}m_{p,j}}{m_{p,i}+m_{p,j}}. & \text{sphere-sphere,} \\ m_{p,j}, & \text{sphere-wall} \end{cases}$$

Again, tangential components are neglected in this work. More details on them can be found in [19].

2.5 Complete four-way coupled fluid-particle interaction algorithm

Taking all information of Sections 2.1 to 2.4 into account results in the complete algorithm of the four-way coupled fluid-particle interaction process, given by Algorithm 1. In order to use a finer resolution in time solely for the simulation of the particle (compare Section 2.4.2), a DEM sub-cycle loop is used. Hence, the particles time step size t_p is defined as

$$\Delta t_p = \frac{\Delta t}{n_{\text{sub}}}$$

with n_{sub} being the total number of sub-cycles, considering the commonly smaller time scale in comparison to the fluid one [9, 10, 45]. Throughout the sub-cycling, the fluid-particle interaction force $\boldsymbol{F}_{p,i}^{fp}$ and torque $\boldsymbol{T}_{p,i}^{fp}$ stay constant.

As a result of inherent fluctuations of the LBM originating from the boundary conditions and its bounce-back nature, an average of $\boldsymbol{F}_{p,i}^{fp}$ and $\boldsymbol{T}_{p,i}^{fp}$ over two-time steps is taken [28].

Algorithm 1: Four-way coupled fluid-particle interaction algorithm (from [19, Algorithm 1])

```
foreach time step t do
    Apply boundary conditions for fluid ⇒ Equation (2.13)
    Calculate fluid-particle interactions ⇒ Equation (2.14) and average over two
      time steps
    Execute LBM step ⇒ Equations (2.4) and (2.5) with collision operator of
      Sections 2.1.1, 2.1.2 and 2.1.4
    foreach DEM sub-cycle do
        Update particle position and rotation ⇒ Equation (2.21a)
        Evaluate lubrication corrections ⇒ Equation (2.17)
        Evaluate collision forces ⇒ Equations (2.22)
        Set external forces ⇒ buoyancy and gravity
        Update particle translational and rational velocity ⇒ Equation (2.21b)
    end
    if necessary then
        Update particle mapping into fluid domain
        Reconstruct PDF information                    } ⇒ Equation (2.15)
    end
end
```

Chapter 3

Stability of Coupled Simulations

In this chapter, we analyze the stability of the four-way coupled algorithm we derived in the last chapter. In order to do that the current limitations of this algorithm are briefly described in Section 3.1. Afterward, in Section 3.2, two possible optimization strategies are discussed, namely increasing the spatial resolution of the simulation domain in Section 3.2.1 and directly optimizing the coupled algorithm at hand in Section 3.2.2. This is followed by a numerical stability analysis in Section 3.3, where at first the term *stability* is defined in the context of this work. Then utilizing multiple test cases, the current state regarding numerical stability is tested with two different simulation setups. The first analyzes the behavior of the coupled algorithm for a sphere moving with a prescribed velocity in Section 3.3.2. The second test case uses a simulation where a sphere can freely settle vertically in Section 3.3.3.

3.1 Current Limitations

The algorithm described in the last chapter can simulate a wide variety of particle-laden flows [19]. Nevertheless, without further optimizations, it is inefficient to simulate and thus analyze turbulent flow problems, e.g., an erosion or evolution of sediment beds. An attempt at such a simulation can be seen in Figure 3.1. The idea for this attempt was the work of Vowinckel *et al.* [1], where erosion events were investigated in a turbulent open channel flow laden with spherical particles. Two particle layers are initialized at the bottom of the simulation domain while the fluid is flowing from left to right. If the fluid velocity is high enough, there is a possibility that a particle is pulled out of such a particle layer. Furthermore, there exists a possibility where moving particles collide with one or more particles in the particle layer and extract them due to this collision. These are exciting phenomena important for understanding erosion events' processes and conditions.

The problem regarding the attempt of Rettinger, seen in Figure 3.1, is that the simulation gets numerically unstable after an unspecified time duration [46] with an acceptable resolution. This issue limits the application of this algorithm immensely because many flow problems in nature are turbulent.

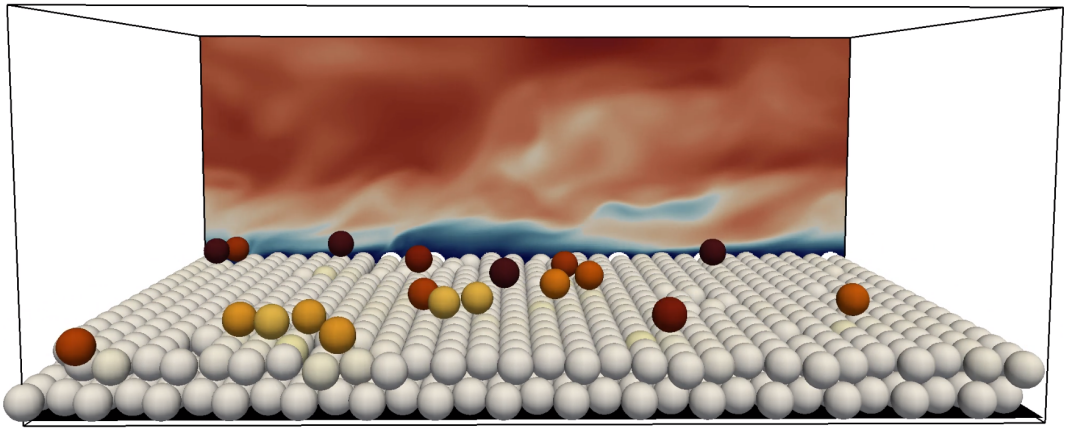


Figure 3.1: Simulation test case of erosion. In the background, the fluid's velocity is shown. Additionally, the color of a particle visualizes its velocity. The darker its color is, the faster is its movement (provided by C. Rettinger).

3.2 Possible Optimization Strategies

We know that the algorithm has problems regarding numerical stability for simulations with high Reynolds numbers. There are multiple ways to overcome this difficulty

- increase the resolution or
- optimize the algorithm.

In the following sections, we will take a closer look at these solution attempts.

3.2.1 Increase the Resolution

Increasing the spatial resolution is straightforward, but understanding why this helps, especially when using the LBM, is not. To fully understand the underlying problem of simply increasing the spatial resolution we introduce a characteristic, dimensionless number that is often used to specify flow conditions for fluid flow simulations and used for comparison of such[1], referring to the Reynolds number (Re) [20]. It is here defined as

$$\mathrm{Re} = \frac{u_c\, l_c}{\nu}. \tag{3.1}$$

with u_c defining the characteristic velocity, l_c the characteristic length, and ν the kinematic viscosity of the fluid. Using the LBM the parameters are given in lattice units. Due to the LBM, the maximal possible fluid velocity we are able to simulate

[1] If the geometry and Reynolds number of two fluid flow simulations are the same, they are the same. This is called the law of symmetry [20].

is $u_c = 0.1$ lattice units. Hence, we have to always ensure that we never reach this velocity and therefore choose small velocities for fluid flow simulations.

To simulate turbulent problems with high Reynolds numbers, we have two ways to achieve high Reynolds numbers, either to increase the spatial resolution and thus the characteristic length l_c or to decrease the kinematic viscosity ν. Often stability problems for LBMs arise from a kinematic viscosity that is chosen too small. Consequently, increasing the spatial resolution is often the only way to increase numerical stability for LBMs.

Especially for DNS' it is impossible without using fine spatial resolutions to resolve everything, where we finally arrive at the problem regarding the further increase of spatial resolution for LBMs. For DNS' with the LBM we already have to use a much finer spatial resolution than physically necessary. If now have to increase this resolution even further, the computational cost increases as a result. With a twice as fine spatial resolution, the computational cost increases 16-fold. Consisting of a factor of 2 for every spatial dimension, hence 2^3 for a three-dimensional problem, and an additional factor of 2 because the physical Δt is halved. As a result the number of time steps for the simulation has to double.

Furthermore, this is not a general solution but an individual for every problem, i.e., if we find a resolution that works for our current problem, it does not mean that it also works for a different problem. Additionally, for simulations, as described in Section 3.1, it is crucial to choose a resolution as coarse as possible but as fine as necessary. Otherwise, the computational resources needed for such a simulation are unnecessarily increased[2] or not feasible at all.

3.2.2 Optimize the Algorithm

In comparison to just increasing the spatial resolution, optimizing a coupled algorithm is much more complex, but at the same time potentially much more rewarding. Particularly, when looking at the possible increase of numerical stability independent of spatial resolution.

Nevertheless, the first step is already intricate, to identify which part of the algorithm can be optimized. The easiest and safest way to detect potential parts to improve is to test each part of the coupled algorithm separately. Unfortunately, this is here and in general often impossible for algorithms with interacting pieces. Consequently, we will use a hierarchy of test cases, where the first simulation setup is relatively easy and gets more complex in the later stages. Initially, we will test as few parts of the coupled algorithm as possible. From then on, if the simulations are stable, we increase the participating parts of the coupled algorithm by using different simulation setups. Like this, it might be possible to detect critical parts regarding numerical stability.

[2] This applies in general, for relatively simple simulations as well as or rather even more for massively parallel simulations executed on supercomputers, where the computational resources are expensive.

For pure flow simulations with the LBM, it is well-known that the collision operator has a significant role in numerical stability. Its influence is well analyzed, and as such, there are advanced variants, like the MRT [15, 16, 47, 48] or cumulant [14, 26, 27] collision operators that significantly increase numerical stability independent of spatial resolution. The same stability increase depending on the choice of the collision operator can be observed for multiphase flow simulations [17, 18, 49]. Nevertheless, the research on the influence of the collision operator on coupled particle-fluid simulations is scarce.

3.3 Numerical Stability Studies

All simulations in this work are implemented with the multiphysics HPC framework waLBerla, which uses block-structured domain decomposition to allow a wide range of applications while adhering to excellent performance and scaling for massively parallel simulations on heterogeneous supercomputers [50]. waLBerla contains modules for different applications; we use

- the LBM-module for the simulation of the fluid flow,
- the MESA-PD module for the particle simulation [51], and
- the coupling module for an explicit coupling of both parts.

In combination with waLBerla, we use the python package *lbmpy*, version 0.3.4, for an automatic code generation for efficient LBM kernels [52], namely the generation of collision operators. Due to its fully symbolic representation of the collision operators, we can easily interchange them using only a few lines of code.

3.3.1 Defining Stability

Due to the complexity of the above-described algorithm, an analytical assessment of stability is not considered feasible here. Instead, we will analyze it regarding its numerical stability, which means we will try different simulation setups to detect where instabilities arise. Therefore, it is essential to define what it means and what conditions have to be fulfilled for a simulation to be *stable* or *unstable*.

- *stable*: A simulation is stable if it runs for 1 000 000 time steps without any values being NaNs. Values here are any velocity, density, forces, etc.
- *unstable*: A simulation is unstable if at any time step in the range of $[0, 1\,000\,000]$ a value is NaN.

3.3.2 Test Case 1 – Sphere Moving with a Prescribed Velocity

Motivation

As explained above, we want to test a few parts of the algorithm as possible to verify or identify possible optimization potential. The probably most straightforward setup would be a pure channel flow without any particles. There, we would be able to extensively analyze the behavior of the fluid solving part, the LBM, but this is already well analyzed, e.g. [25]. The following more complex setup features a fixed sphere in the simulation domain. Here we could, in addition to the LBM, analyze the CLI boundary condition of the explicit particles, but this is also already well-tested [30]. Therefore the first test case we will run features a particle in motion with a prescribed velocity in a cuboid channel filled with fluid. The setup is shown in Figure 3.2 and analyzes the following three parts of the algorithm:

- the CLI BC of the particle,
- fluid flow solver ⇒ LBM with collision operator of Section 2.1.3, and
- re-initialization of the cells transitioning from solid to fluid state.

Description

The simulation features a cuboid channel with a size of $w \times w \times h = [4 \times 4 \times 8]d_p$, no-slip boundary conditions for the boundaries in x and y-direction, while periodic boundary conditions are used in the z-direction. In addition, an explicitly mapped sphere with a diameter of $d_p = 10$ cells is initialized in the center of the simulation domain. We intentionally use a rather small diameter for the particle because we are interested in increasing the stability of this algorithm, and we know that with an increased resolution, we gain better stability. The sphere is moving with a prescribed velocity $\boldsymbol{u}_{ppv} = (0, 0, 0.2)^{\mathrm{T}}$ lattice units. In order to prevent a sudden disruption of the fluid field at the beginning of the simulation, we artificially accelerate it according to

$$\boldsymbol{u}_p(t) = -\boldsymbol{u}_{ppv}\left(1 - \exp\left(-c_{\mathrm{acc}}\frac{t}{t_{\mathrm{ref}}}\right)\right) \tag{3.2}$$

with t_{ref} being the particles reference time, computed with

$$t_{\mathrm{ref}} = \frac{d_p}{\|\boldsymbol{u}_{ppv}\|}.$$

For the specification of the flow conditions, we use the Reynolds number defined in Equation (3.1) where $u_c = \|\boldsymbol{u}_{ppv}\|$ and $l_c = d_p$.

As described above, the simulation is run for $T = 1\,000\,000$ which is equal to $T = 2000t_{\mathrm{ref}}$ and results in approximately 250 cycles of the particle through the simulation domain.

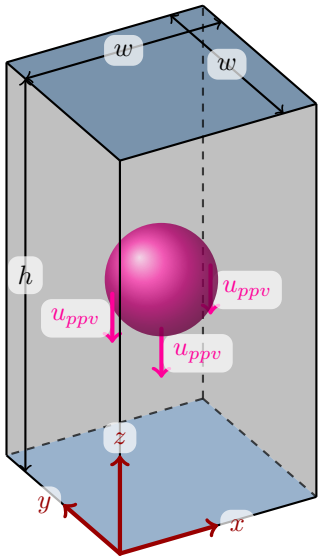


Figure 3.2: Sphere moving with prescribed velocity $\boldsymbol{u}_{ppv} = (0, 0, -0.02)^{\mathrm{T}}$ through a cuboid channel with periodic BCs in z-direction‘

Results

Besides the standard D3Q19 TRT collision operator the two variants with increased bulk viscosity are tested because the work of Rettinger [19] mainly analyzed problems with moderate Reynolds numbers. Hence, we will check if the observed feature of decreasing nonphysical oscillations is also present in the case of high Reynolds numbers and turbulent flows.

With the standard TRT collision operator, it is possible to run this setup with a Reynolds number up to 1081, while the other variants got unstable with much lower Reynolds numbers. For example, while the TRT+aB model could run stable simulations with a maximal Reynolds number of 500, the TRT+B model could only run this test case with a Reynolds number of 170.

We conclude that increasing the bulk viscosity with either of the two ways described before the simulation with this setup results in less stable conditions for problems with high Reynolds numbers. Khirevich *et al.* reported in [25] that high values for Λ_b can lead to instabilities. Therefore we will continue the limitation testing with the standard TRT collision operator.

3.3.3 Test Case 2 – Vertically Freely Settling Sphere

Motivation

Now it is time to test a fluid-particle coupling, where particles and fluid interact, which means we are now adding the MEM. Due to the fixing of the sphere or prescribing the particle's velocity, only the fluid was influenced by the particle but not vice versa. To achieve that, we let a gravitational force act on the particle. We will fix the particle in x and y-direction in the middle of the simulation domain, such that it can only move freely in z-direction. The reason for this setup is to prevent collisions with the boundaries that might occur due to a possible turbulent flow field that might horizontally displace the sphere. We will investigate collision behavior in more detail in Chapter 4.

Description

As in Section 3.3.2, the simulation domain is bounded by no-slip BCs in x and y-direction, and periodic BCs in z-direction. The setup for a settling sphere in a cuboid filled with a fluid is shown in Figure 3.3.

For this test case, we cannot use the Reynolds number for characterization because it is not known at the beginning of the simulation. Instead, we will use the Galileo number Ga, which quantifies the importance of gravity and is given by

$$\mathrm{Ga} = \sqrt{\frac{\Delta\rho}{\rho_f}\frac{g d_p^3}{\nu^2}}$$

where $\Delta\rho = \rho_f - \rho_p$ and g describes the magnitude of the gravitational acceleration [30]. Using a density ratio $\rho_p/\rho_f = 2$ together with the gravitational velocity u_g defined by

$$u_g = \sqrt{\frac{\Delta\rho}{\rho_f} d_p g}$$

which is set to $u_g = 0.02$ lattice units, we can to compute all necessary parameters for our simulation. Here we will increase the Galileo number as much as possible, until the simulation gets unstable. In order to compare this test case with the setup from before, we have to calculate the Reynolds number after the simulation is finished. Hence, we compute the mean particle velocity of the last 50 000 time steps and use this as the characteristic velocity u_c in Equation (3.1).

Results

We could simulate this setup for a million time steps up to a Galileo number of $\mathrm{Ga} = 250$. The resulting particle velocities in z-direction for the stable simulation with

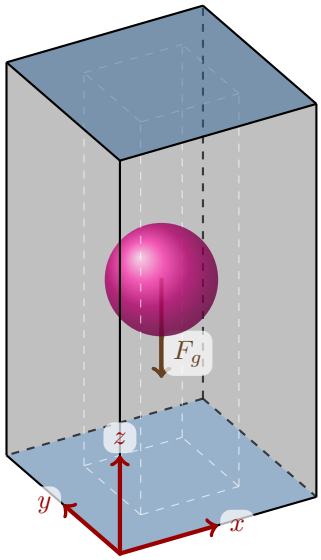


Figure 3.3: Horizontally fixed sphere (indicated with the white dashed cuboid) settling in a cuboid channel due to gravity

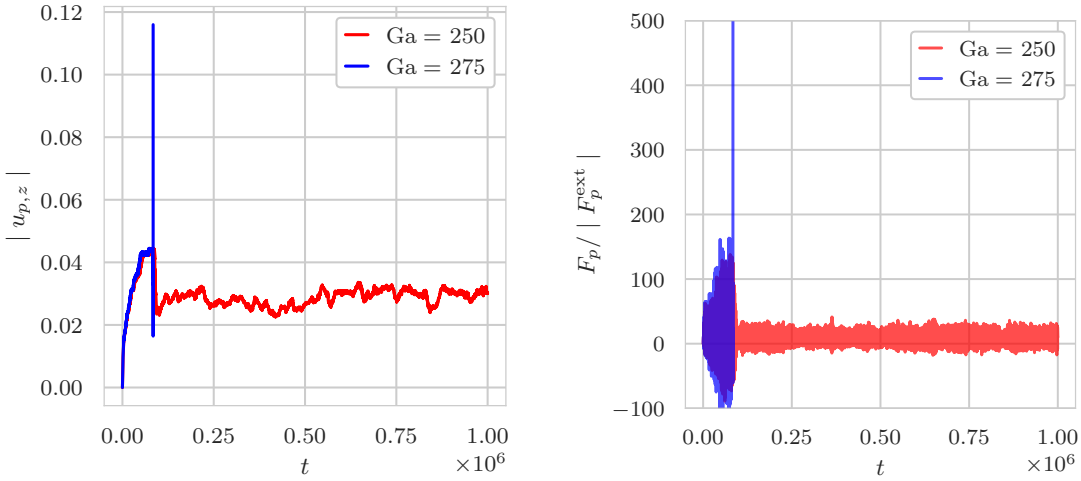
Ga = 250 and an unstable setup with Ga = 275 are visualized in Figure 3.4a. Here it can be observed that the particle accelerates for both cases until the flow field gets turbulent. Hence, the particle velocity drops and afterward stabilizes around a velocity of $u_{p,z} = 0.025$ lattice units for a Galileo number of Ga = 250. The unstable setup also features the drop in particle velocity directly after the flow field gets turbulent but then increases to a value of almost $u_{p,z} = 0.12$. This sudden change of the particle's behavior can be explained by the acting normalized hydrodynamic interaction force during the simulation shown in Figure 3.4a, which is exploding to a value of over $8e^{63}$.

In order to compare this setup with the previous one, we take the average of the velocity magnitude of the last 50 000 time steps resulting in $\bar{u} = 0.029968$, with which we can compute the Reynolds number of this setup according to Equation (3.1) Re = 374.60. This is much lower compared to the previous test. The interaction of flow and sphere motion results in a more complex fluid flow field. Hence it is difficult to identify the part inserting the numerical instability.

3.3.4 Extended Studies

As described in Section 3.2.2, it is well-known that for pure flow simulations, the collision operator has a significant role regarding numerical stability. Taking this into consideration and the advantage of using *lbmpy*, we can easily change collision

[3] The y-axis of Figure 3.4b was limited to the seen range, otherwise one would have only visible a red zero-line and a blue graph having the value zero until the sudden increase. Consequently omitting the behavior of the hydrodynamic interaction force during the simulation.



(a) Particle velocity in z-direction

(b) Hydrodynamic interaction force

Figure 3.4: Simulation results for a stable and unstable setup of a vertically freely settling sphere

operators with only a few lines of code. Therefore, we will now test the cumulants, which can simulate fluid flow problems with very high Reynolds numbers already shown by Geier *et al.* [14, 26, 27]. However, it has to be noted that this collision operator was not yet used for coupled particulate flows.

Version 0.3.4 of *lbmpy* already offers the cumulant collision operator with different stencils. We will test both the D3Q27 as well as the D3Q19 stencil, with the following relaxation rates respectively:

- D3Q19 $\Rightarrow \{0, 0, 0, 0, s_v, s_v, s_v, s_v, s_v, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1\}$
- D3Q27 $\Rightarrow \{0, 0, 0, 0, s_v, s_v, s_v, s_v, s_v, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1\}$

Using the D3Q19 stencil results in even more unstable simulations, such that we were not able to simulate the given setup with a Galileo number of 250. Nevertheless, with the D3Q27 and the above-defined relaxation rates, we could simulate test cases with a Galileo number of 10 000, resulting in a Reynolds number of $\approx 1.3e^4$.

With this, we can be confident that the other involved parts of the coupled algorithm, like the CLI boundary conditions and the refilling of the transitioning cells from solid to fluid cells, are sufficient and can also be used in simulations with high Reynolds numbers.

3.4 Summary

In this chapter, we analyzed the current state and limitations of the coupled algorithm developed by Rettinger in [19, 30]. We showed that an increasing bulk viscosity for particulate flows with high Reynolds numbers results in decreasing stability. Furthermore, the collision operator is shown to have a a significant impact on stability in turbulent particulate flows.

In addition to that, we could confirm that it is really difficult or rather almost impossible to test specific parts of a coupled algorithm with simulation setups where these parts are not tested separately.

With this knowledge, we will physically validate different collision operators before analyzing the numerical stability. The motivation for this shift is that algorithmic changes always have to be accompanied by validation studies to ensure ongoing correctness. Otherwise, one might end up with a stable but inaccurate simulation that cannot be used for any study.

In addition to the collision operators presented so far, we will analyze the default D3Q19 MRT collision operator used by *lbmpy* derived in [52] and the D3Q27 MRT collision operator used by Peng et al. for the simulation of particulate turbulent flows [53–55]. Here an overview of all collision operators and how they will be represented starting from the next chapter on:

- TRT-C ⇒ incompressible TRT-BB collision operator with a bulk viscosity rate factor of 1, as defined in Section 2.1.3
- Cumulants ⇒ compressible cumulant collision operator from Geier *et al.* derived in [14]
- MRT-L ⇒ incompressible MRT collision operator derived from Bauer with *lbmpy* in [52]
- MRT-P ⇒ compressible MRT collision operator developed by Peng *et al.* in [53–55]

Further information on the given collision operators, like the moment matrices or equilibrium values, can be found in Appendix A.

Chapter 4

Influence of the LBM Collision Operator on Particle Interaction

We showed in the last chapter that the collision operator plays a vital role concerning numerical stability for particular flows. Consequently we physically validate several collision operators with a test setup, where a sphere is settling freely under gravity until it collides with the bottom wall. The resulting simulation data, including the gap size trajectories of the bottom surface of the sphere and the bottom wall, are investigated and compared against measurement data gained from experiments reported by Gondret *et al.* in [56].

4.1 Motivation

For the physical validation, we use a wet sphere-wall-collision setup [19], shown in Figure 4.1. This simulation features a settling sphere, which eventually collides with the bottom wall of the domain. All parts of the four-way coupled algorithm described in Chapter 2 are involved.

The characteristic number representing this test case is the Stokes number St, which describes the ratio of the particle's inertia to the viscous forces [56] and is given by

$$\mathrm{St} = \frac{1}{9}\frac{\rho_p}{\rho_f}\frac{u_s d_p}{\nu_f},$$

with u_s defining the particle's impact velocity. Following [19], we will use the rebound trajectory and height of the particle for validating the different collision operators and compare these results with the experimentally determined measurements of Gondret *et al.* [56].

4.2 Description

For this test case, we are using

- periodic boundary conditions for the side walls in x and y-direction,

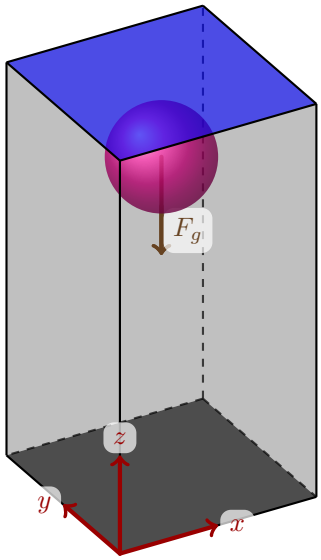


Figure 4.1: Setup of sphere-wall collision test case.

- no-slip boundary conditions for the bottom wall in negative z-direction, and
- outflow boundary conditions at the top of the box in positive z-direction.

The settling sphere is initialized $3/2 d_p$ cells below the top of the simulation domain and horizontally centered with a diameter that will be defined later on. To ensure that the sphere reaches its terminal velocity of 0.02 lattice units and we again use an artificial acceleration as described in Section 3.3.2 Equation (3.2), except for using t_{St} instead of t_{ref} defined as

$$t_{\mathrm{St}} = \frac{\rho_p}{\rho_f} \frac{d_p^2}{18\nu_f}.$$

Here, the same parameters are used as in [19]; the acceleration constant c_{acc} is set to 10, and the density ratio ρ_p/ρ_f. Once the distance between sphere and wall is less than the diameter of the sphere d_p, the artificial acceleration is turned off, and the particle settles freely under the acting gravitational force and buoyancy, where we use a gravitational acceleration of $g = 9.81\ \mathrm{m/s^2}$.

Furthermore, we use initially the following parameters tuned and validated by Rettinger in [19]:

- collision time $T_c = 2.31\, d_p/c_s$
- subcycle number of rigid body simulation $n_{\mathrm{sub}} = 100$
- minimal lubrication gap size $\delta_{n,\mathrm{min}}^{\mathrm{lub}}$ according to

$$\delta_{n,\mathrm{min}}^{\mathrm{lub}} = \left(1e^{-3} + 7e^{-5}\frac{d_p}{2}\right)\frac{d_p}{2}$$

Table 4.1: Sphere-wall-collision parameters used in the experiment by Gondret *et al.* in [56] and for the validation of parameters for the four-way coupled algorithm derived by Rettinger in [19].

St	Re	ρ_f [kg/m³]	μ_f [Ns/m²]	d_p [m]	ρ_p [kg/m³]	domain size [d_p]
27	30	965	0.1	0.006	7800	$12 \times 12 \times 48$
100	110	953	0.02	0.004	7800	$12 \times 12 \times 48$
152	164	935	0.01	0.003	7800	$12 \times 12 \times 64$

Table 4.1 contains additional simulation parameters used in experiments by Gondret *et al.* in [56] and by Rettinger in [19] for determining the above-mentioned feasible parameters for the four-way coupled algorithm.

4.3 Results

First, we compare the trajectories resulting from the gap sizes between the bottom wall and the lower surface of the sphere with a diameter of $d_p = 20$ cells for the test case with a Stokes number of 152, in Figure 4.2. The trajectories are shifted such that the first impact time is set to zero. We observe a different collision behavior for the various collision operators. The TRT-C version is closest to the measurements from Gondret *et al.* [56], visualized as a red and black graph, respectively. The cumulant and MRT-L collision operators show a similar collision behavior, with a much smaller gap size for the first bounce and an almost non-existent second and third bounce. The graph for MRT-P is missing because the simulation got unstable after a few thousand-time steps. The reason for that was the forces being NaN. From this, we can conclude that the collision operators in the current version of the LBM in the four-way coupled algorithm are not usable, either because of insufficient representation of the collision behavior physics or stability.

With such a coupled system, there are multiple possibilities where these differences regarding the collision behavior can arise from. For further investigation, the acting forces on the particle are plotted with the normalized velocity development (with $u_{\text{ref}} = 0.02$ lattice units) in z-direction against the shifted normalized time (with $t_{\text{ref}} = {}^{d_p}/_{u_{\text{ref}}}$) for the three collision operator variants, shown in Figure 4.3. The two gray vertical dotted lines show the start and end of the collision, respectively.

Before comparing the three variants and discussing the differences, the graphs for TRT-C, shown in Figure 4.3a, are briefly described. Right before the collision starts, indicated with the first vertical dotted gray line, the lubrication correction force $F_p^{\text{lub-corr}}$ with a decreasing distance between sphere and wall. Consequently, the velocity is decreasing slightly. The moment the sphere and wall are in contact, the lubrication

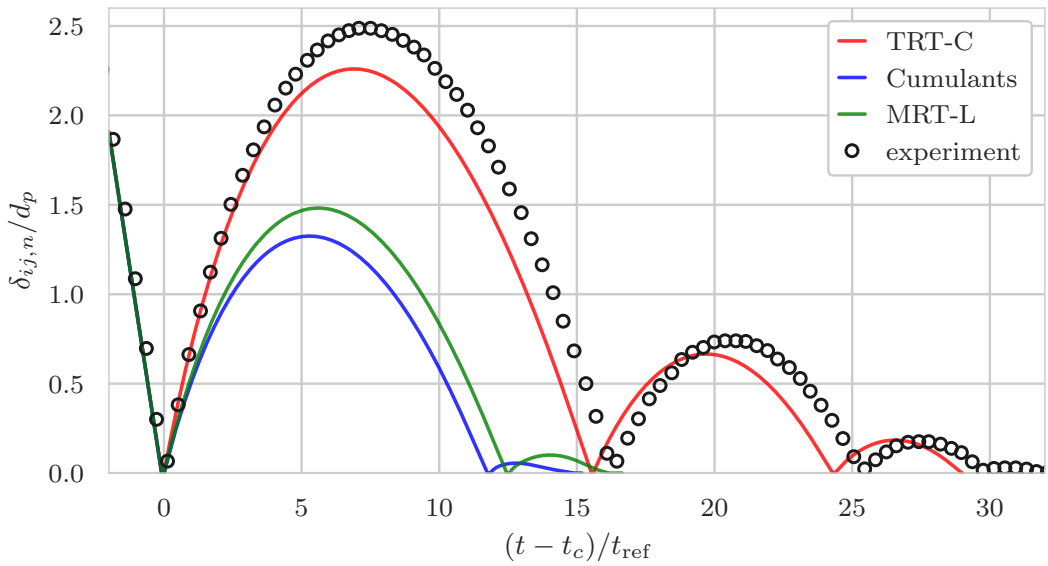
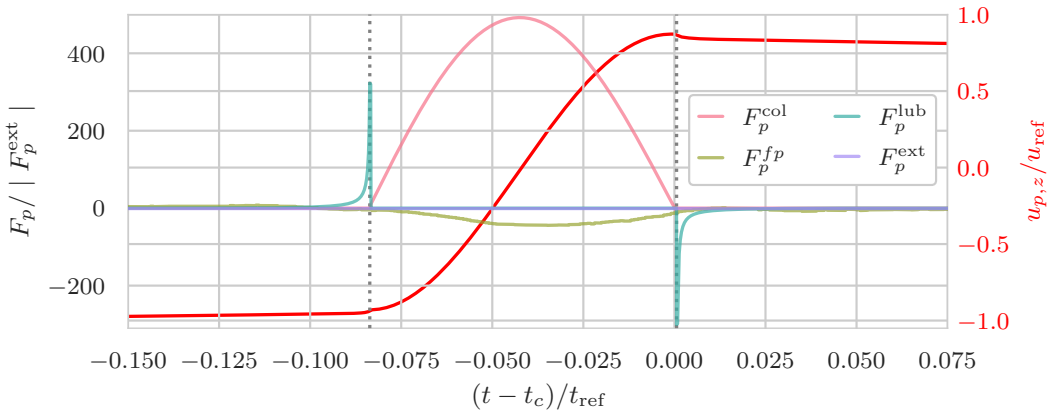


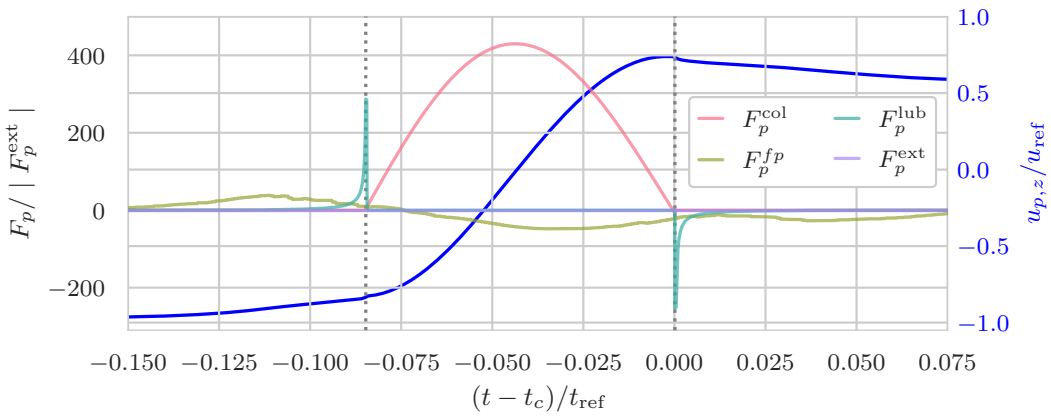
Figure 4.2: Trajectories of normalized particle-wall gap sizes over shifted normalized time for different collision operators and a measurement data from Gondret *et al.* [56] with a Stokes number of 152.

correction force F_p^{lub} is set to 0. Contrarily the collision force starts to act, resulting in a further decreasing particle velocity. F_p^{col} increases until the particle's velocity hits 0, then it decreases up to the end of the collision, visualized with the second vertical dotted gray line.

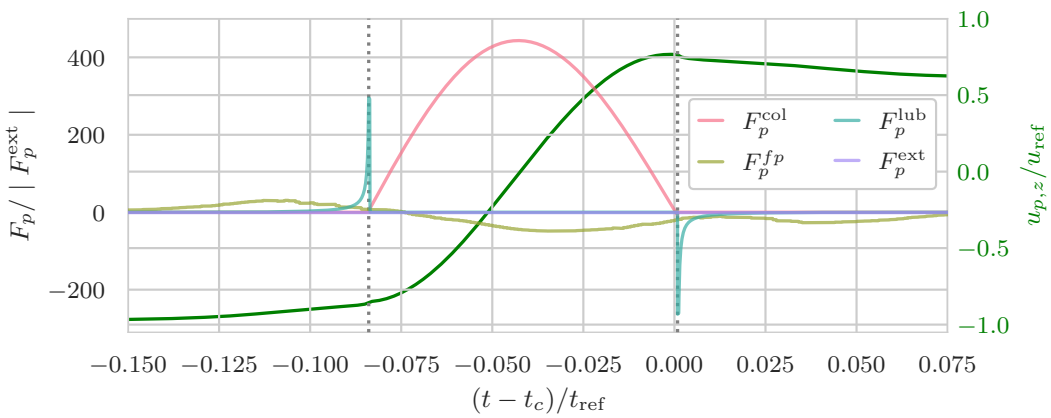
In the following we will investigate the other two variants, and analyze where the different collision behavior originates from. Both the cumulants, see Figure 4.3b, and MRT-L in Figure 4.3c are similar, which is also reflected in the trajectories of the gap sizes, cf. Figure 4.2. Opposite to the TRT-C collision operator, these two show an increasing hydrodynamic interaction force F_p^{fp} before the collision or lubrication correction even starts. The same phenomenon is observed for the TRT-C method, but to a small degree. For easier comparison, the hydrodynamic interaction forces of the three collision operators are plotted together in Figure 4.4. Here, F_p^{fp} is approximately $7 - 8$ times larger for the cumulants and MRT-L than for the TRT-C variant. This results in a sequence of other deviations like a decreasing velocity, such that the effective collision velocity of the particle is, cf. Figure 4.5b. Additionally, the particle velocity development over the entire simulation time is shown in Figure 4.5a, where the terminal velocity of $u_{\mathrm{ref}} = 0.02$ lattice units is reached for all collision operators. As a consequence of the lower collision particle velocity the lubrication correction force F_p^{lub} has a lower peak as well as the collision force F_p^{col}. Even after the end of the collision, the absolute hydrodynamic interaction force F_p^{fp} is bigger than the one for



(a) TRT-C



(b) Cumulants



(c) MRT-L

Figure 4.3: Acting forces on particle and resulting particle velocity before, during, and after collision for Stokes number St = 152. Vertical gray dotted lines mark beginning and end of sphere-wall collision.

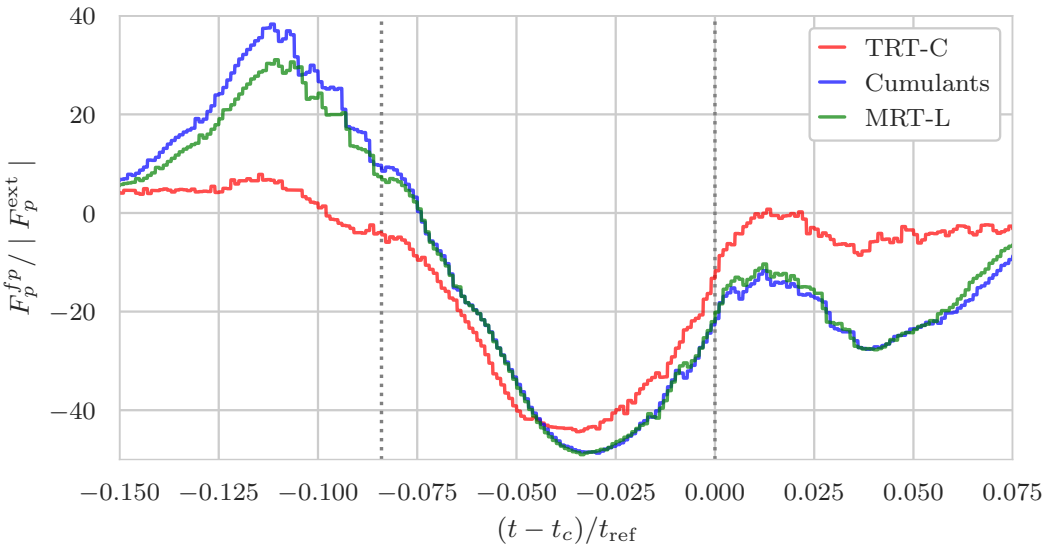


Figure 4.4: Hydrodynamic interaction forces before, during, and after the collision for TRT-C, cumulants and MRT-L for St = 152. The vertical gray dotted lines represent beginning and end of the sphere-wall collision.

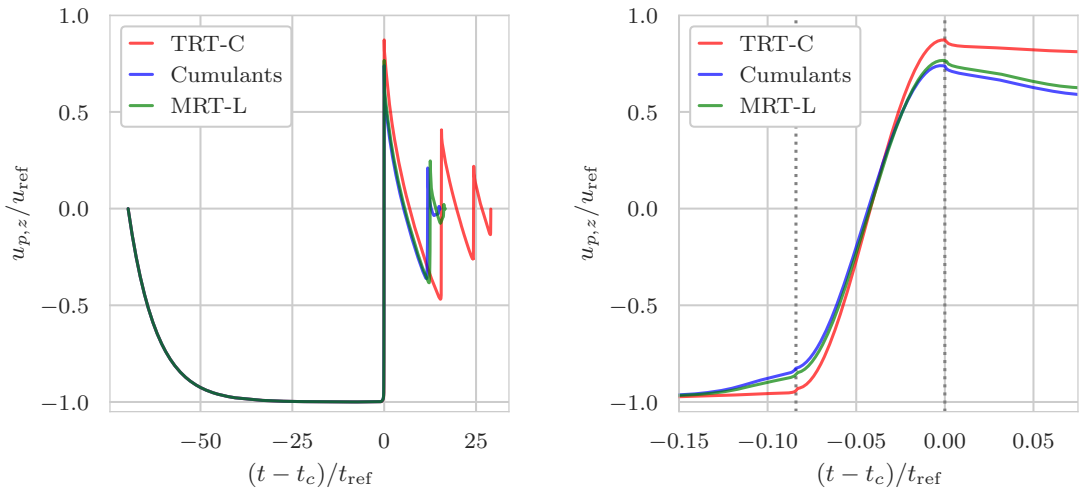
the TRT-C, which again results in a decreasing particle velocity. This explains that the second and third bounces of the particle are almost non-existent, cf. Figure 4.2.

For now, let us continue with the analysis of the different collision behavior solely between the TRT-C and MRT-L collision operators, as the cumulants act in a different collision space.

4.3.1 Comparison of TRT-C and MRT-L

The difference between these collision operators is the different used moments and relaxation rates. By changing the relaxation rates of the MRT-L collision operator to a TRT parameterization (called now TRT-L), we can check if the varying setting of the relaxation rates is responsible for the diverging collision behavior.

The resulting hydrodynamic interaction force before, during, and after the collision can be seen in Figure 4.6, as well as the trajectories of the gap sizes for both TRT-C and TRT-L collision operators. As we can see, both variants have identical results. This is surprising because, regarding pure fluid flow simulations with the LBM, only specific relaxation rates influence the physics, as mentioned in Section 2.1.1. These are precisely the relaxation rates up to the 9th moment, which is identical for the TRT-C and unmodified MRT-L collision operator. The other relaxation rates should only influence accuracy and stability.



(a) Complete simulation

(b) Before, during, and after collision

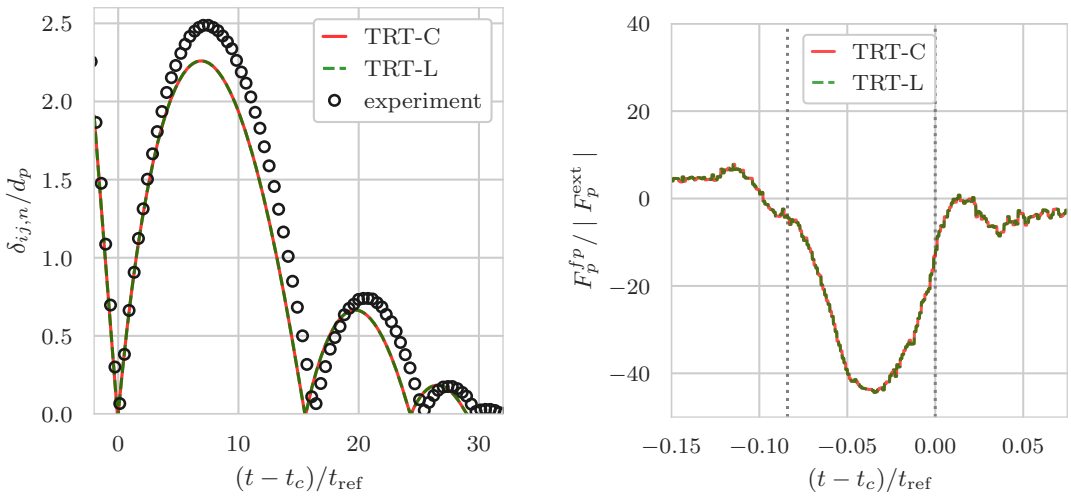
Figure 4.5: Normalized particle velocity development for different time sections with St = 152

Hence, as an intermediate outcome, we can record that the relaxation rates significantly influence the collision behavior within this coupled algorithm for the simulation of particle-laden flows by affecting the hydrodynamic interaction force. Consequently, we will further analyze the influence of relaxation rates on the collision behavior in a sphere-wall collision setup with a parameter study.

4.3.2 Parameter Study regarding the Relaxation Rates for MRT-L

By default, many simulations have to be executed during a parameter study. Therefore it is crucial to reduce the computational cost per simulation as much as possible, in this case, by reducing the sphere diameter from $d_p = 20$ to 10 cells. Even with this reduced resolution, the simulations are still stable. However, the overall accuracy is reduced, but the tendency persists, which is shown in Figure 4.7. The gap size trajectory for the TRT-L collision operator is still significantly closer to the measured data with $d_p = 10$ than the data resulting from the MRT-L collision operator with the same particle diameter.

In addition, we reduce the number of collisions from four to two, which is why the collision operator versions with $d_p = 10$ offer only one bounce in Figure 4.7. The motivation here is the further reduction of the computational cost and the expressiveness of the first bounce. If the first bounce deviates significantly from the expected outcome, the following bounces will do the same. Therefore, we will compare most of the



(a) Gap size trajectories **(b)** Hydrodynamic interaction force

Figure 4.6: Comparison of TRT-C and TRT-L collision operator w.r.t. the hydrodynamic interaction force and gap size trajectories for St = 152

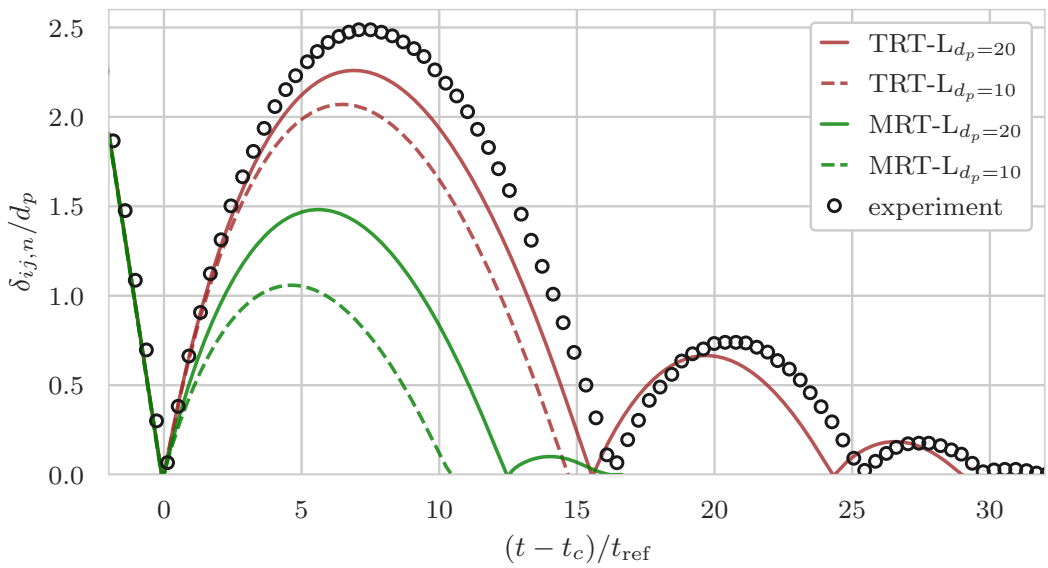
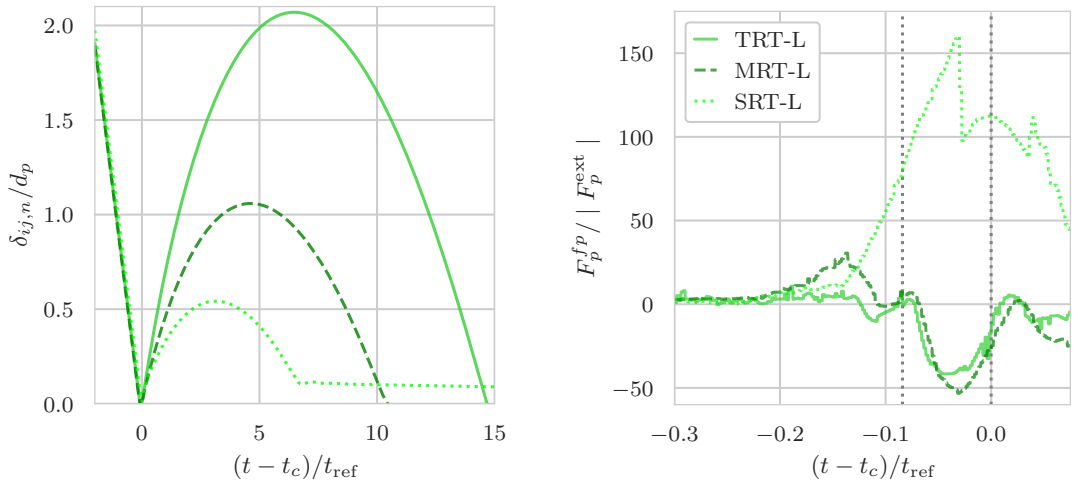


Figure 4.7: Gap size trajectories for St = 152 for the MRT-L and TRT-L collision operators with a particle diameter of $d_p = 10$ or 20



(a) Hydrodynamic interaction force

(b) Gap size trajectories

Figure 4.8: Comparison of TRT-L, MRT-L, and SRT-L collision operators with regard to gap size trajectories and the normalized hydrodynamic interaction force for a Stokes number of St = 152

analyzed different relaxation rate sets concerning the rebound height H_r according to

$$H_r = \max_t \left(\frac{\delta_{ij,n}(t)}{d_p} \right), \qquad \text{with } t > t_c.$$

We will exclude the sets, where unexpected phenomena are occurring. The rebound height H_r describes the biggest gap size between the lower surface of the particle and the wall after the first collision.

As we have already analyzed the standard TRT parameterization with the MRT-L collision operator, we will also look at the SRT parameterization. As a reminder for an SRT collision operator, all relaxation rates, except the first four, are set to s_ν. The resulting gap size trajectories, as well as the normalized acting hydrodynamic interaction forces on the sphere, are plotted in Figure 4.8a and Figure 4.8b respectively for the SRT-L and as comparison a likewise for TRT-L and MRT-L.

Here, the collision behavior for the SRT deviates even more from the expected one because it seems to not collide with the wall. The minimum resulting gap size between particle and wall for the first "collision" is 0.04 cells. Looking at Figure 4.8b, we see that the evolution of the hydrodynamic interaction force is completely different from the development of the other two variants. Hence, we will take a closer look at what is happening during the "collision" for the SRT collision operator. The acting forces and the resulting development of the particle velocity are plotted in Figure 4.9 for a section around the time of the "collision". We can assume that there is no collision taking place. To prove this hypothesis, the lubrication correction force and the collision force

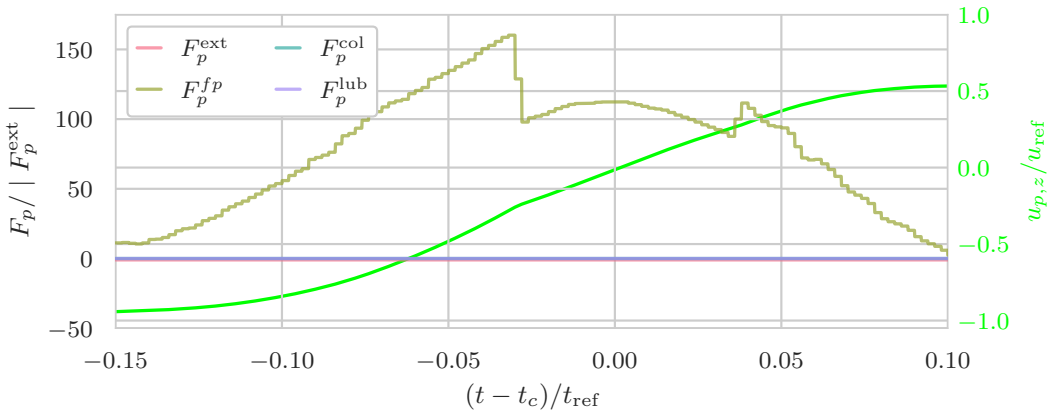


Figure 4.9: Acting forces on particle and resulting particle velocity before, during, and after the "collision" for the SRT-L collision operator.

are exclusively visualized together with the resulting particle velocity. Figure 4.10 shows the expected absence of the collision. The particle gets so close to the wall that the lubrication correction model is activated but only a few moments later already switched off again, while the collision force is zero during the whole process. The hydrodynamic interaction force increases so rapidly before the collision that the particle comes to rest and is reflected before this action can happen. Consequently, comparing the hydrodynamic interaction forces between SRT/TRT/MRT-L collision operators is deceiving, because the time period where the collision is taking place does not exist for the SRT-L method. Furthermore, the simulation data for the SRT-L in Figure 4.8b is shifted according to $(t - t_c)$, although there is no collision happening. t_c is here defined as the time the particle's velocity switches sign.

Hence, we can record that via a particular setting of the relaxation rates, entire physical processes may be altered during a collision. Taking the SRT parameterization as an example, we see that neither the lubrication correction model is activated nor the discrete element method. As such, a proper collision treatment is not possible. This also concludes that looking exclusively at the rebound height H_r as an evaluation tool is insufficient. If not otherwise stated, the following rebound heights of the analyzed parameter setups result from an actual collision, where particle and wall are in contact during the collision process, not as it is in the case of the SRT collision operator.

In the upcoming analysis, we will use the following composition for the relaxation rates:

$$\boldsymbol{S}^{\text{MRT-L}_{\text{PS}}} = \text{diag}(0,\, 0,\, 0,\, 0,\, s_\nu,\, s_\nu,\, s_\nu,\, s_\nu,\, s_\nu,\, \omega^*,\, \omega_0,\, \omega_1,\, \omega_2,\, \omega_3,\, \omega_4,\, \omega_5,\, \omega_6,\, \omega_7,\, \omega_8).$$

As described in Section 2.1.1, the first four relaxation rates equal to zero, so that the conservation of mass and momentum is satisfied. ω^* represents the relaxation rate that controls the bulk viscosity. Additionally, the 5th to 9th relaxation rates are also

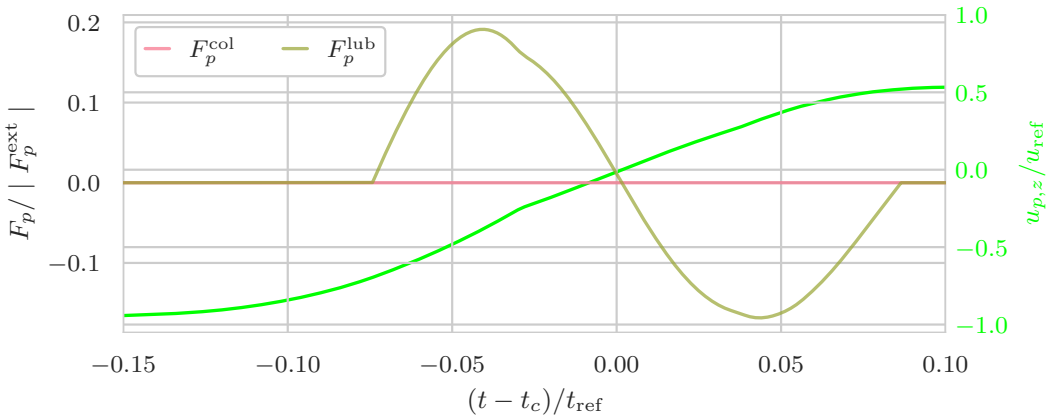


Figure 4.10: Development of lubrication correction force and collision force around the time of the "collision".

automatically determined according to Equation (2.9) in order to simulate a fluid with given viscosity.

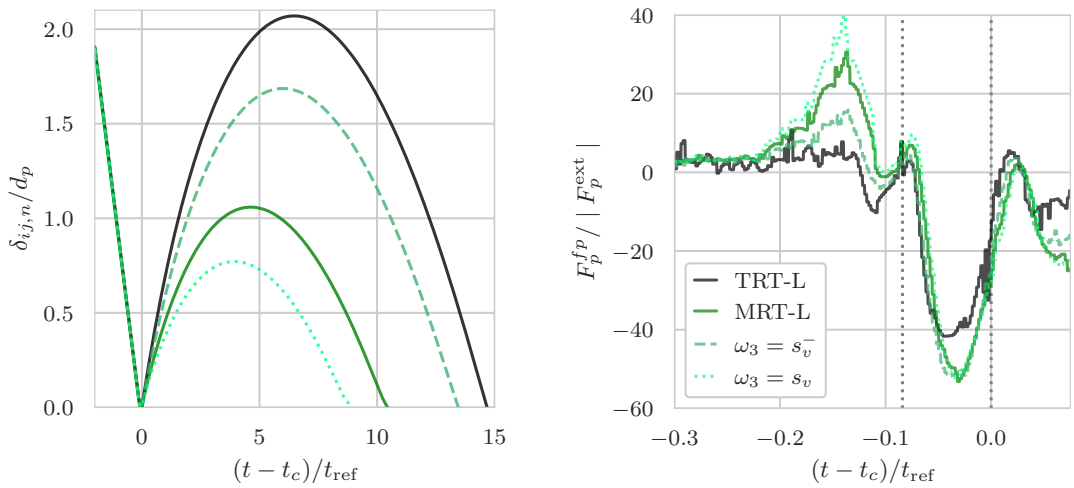
First, we will start the parameter study by analyzing how modifying single relaxation rates changes the collision behavior. For this, we will use the MRT-L collision operator and change one relaxation rate either to s_v according to Equation (2.9), resulting in $s_v = 1.98547$ or s_v^- by using Equation (2.12) with $\Lambda = 3/16$ resulting in $s_v^- = 0.0193237$. With these values, we cover a vast range, where s_v is close to the maximal and s_v^- to the minimal possible relaxation rate $\in (0, 2)$, while $\omega_x = 1$ for $x \in [0, 8]$ is already tested with the unmodified MRT-L.

Table 4.2 shows the resulting rebound height H_r for the described parameter sets. Choosing the relaxation rates as one would for the TRT parameterization, H_r gets closer to the measured data from Gondret *et al.* [56]. This means that setting the relaxation rates $\omega_x = s_v^-$ for $x \in [0, 5]$ and $\omega_x = s_v$ for $x \in [6, 8]$ shifts the resulting rebound height in the right direction, while setting them in the opposite damps the collision behavior even further, which is exemplified in Figure 4.11. Here, the impact of setting ω_3 is directly visualized. As expected from the gained knowledge up to this point, it directly influences the behavior over the hydrodynamic interaction force before the collision is executed. Additionally, setting specific relaxation rates like ω_3 and ω_5 has a greater influence than setting others. Nevertheless, the gap size trajectories of the TRT-L are not reached by setting only one relaxation rate different from the unmodified MRT-L.

Taking a closer look at the rebound height presented in Table 4.2, setting, e.g., ω_0 and ω_2, ω_1 and ω_4, or ω_3 and ω_5 in the described way results in the same rebound height. Therefore, we tried to set both or even four of them to s_v^- which is shown in Table 4.3. Here, we got interesting results, while the combination of $\omega_0 = \omega_2$, and $\omega_1 = \omega_4$ resulted in a slightly or much higher rebound height, respectively. Setting

Table 4.2: Evaluation regarding the rebound height H_r for various sets of relaxation rates. The rebound height of all parameter sets closer to the measurement data than the MRT-L collision operator are highlighted bold.

ω^*	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8	H_r
				TRT-L						2.07
				MRT-L						1.06
s_ν	s_ν^-	1	1	1	1	1	1	1	1	**1.45**
s_ν	1	s_ν^-	1	1	1	1	1	1	1	**1.19**
s_ν	1	1	s_ν^-	1	1	1	1	1	1	**1.45**
s_ν	1	1	1	s_ν^-	1	1	1	1	1	**1.69**
s_ν	1	1	1	1	s_ν^-	1	1	1	1	**1.19**
s_ν	1	1	1	1	1	s_ν^-	1	1	1	**1.69**
s_ν	1	1	1	1	1	1	s_ν^-	1	1	0.928
s_ν	1	1	1	1	1	1	1	s_ν^-	1	0.828
s_ν	1	1	1	1	1	1	1	1	s_ν^-	0.791
s_ν	s_ν	1	1	1	1	1	1	1	1	0.972
s_ν	1	s_ν	1	1	1	1	1	1	1	0.769
s_ν	1	1	s_ν	1	1	1	1	1	1	0.972
s_ν	1	1	1	s_ν	1	1	1	1	1	0.772
s_ν	1	1	1	1	s_ν	1	1	1	1	0.601
s_ν	1	1	1	1	1	s_ν	1	1	1	0.772
s_ν	1	1	1	1	1	1	s_ν	1	1	**1.30**
s_ν	1	1	1	1	1	1	1	s_ν	1	**1.18**
s_ν	1	1	1	1	1	1	1	1	s_ν	**1.15**



(a) Hydrodynamic interaction force

(b) Gap size trajectories

Figure 4.11: Comparison of different relaxation rate parameterization for ω_3 with regard to gap size trajectories and the normalized hydrodynamic interaction force for a Stokes number of St = 152

$\omega_3 = \omega_5$ produced a smaller rebound height than for setting only one of these to s_v^-. Furthermore, combining two of those relaxation rate sets yielded interesting results again. Choosing $\omega_0 = \omega_2 = \omega_1 = \omega_4$ produced approximately the mean height of both separate parameterization. Another set $\omega_0 = \omega_2 = \omega_3 = \omega_5$ made the simulation unstable and the combination of $\omega_1 = \omega_4 = \omega_3 = \omega_5$ increased the rebound height significantly taking it almost up to the TRT-C variant, although one pair was damping and the other one amplifying H_r. These results show that no trivial way of predicting how one how one combination modifies the collision behavior exists.

Lastly, we are left with combining all three pairs of relaxation rates, such that $\omega_{0-5} = s_v^-$. We set all relaxation rates of the third-order moments to agree with a TRT parameterization[1]. As such, we directly tested setting $\omega_{6-8} = s_v$, following the TRT parameterization one would set the relaxation rates of even-odd order moments in this way. The resulting rebound heights of both parameter variants are shown in Table 4.4. Here, the first version results in an even higher H_r while the second version is influencing H_r only slightly.

Now that we found a method where the resulting rebound height is closer to the measurement data, the question arises if the rebound height can be further increased to simulate the collision behavior as accurately as possible. Therefore, Table 4.5 shows the rebound heights for $\omega_{0-5} = s_v^-$ and the possible combinations for ω_x for $x \in [6, 7, 8]$.

[1]The TRT parameterization sets the relaxation rates of odd order moments to s_v^-

Table 4.3: Rebound height H_r for different relaxation rate combinations set to s_v^- for St = 152.

ω^*	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8	H_r
s_ν	s_ν^-	1	s_ν^-	1	1	1	1	1	1	1.60
s_ν	1	s_ν^-	1	1	s_ν^-	1	1	1	1	1.76
s_ν	1	1	1	s_ν^-	1	s_ν^-	1	1	1	1.31
s_ν	s_ν^-	s_ν^-	s_ν^-	1	s_ν^-	1	1	1	1	1.69
s_ν	s_ν^-	1	s_ν^-	s_ν^-	1	s_ν^-	1	1	1	—
s_ν	1	s_ν^-	1	s_ν^-	s_ν^-	s_ν^-	1	1	1	1.95

Table 4.4: Rebound height H_r for different relaxation rate combinations for St = 152.

ω^*	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8	H_r
s_ν	s_ν^-	s_ν^-	s_ν^-	s_ν^-	s_ν^-	s_ν^-	1	1	1	2.17
s_ν	1	1	1	1	1	1	s_ν	s_ν	s_ν	1.35

Table 4.5: Rebound height H_r for different relaxation rate combinations for St = 152.

ω^*	ω_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8	H_r
s_ν			s_ν^-				s_ν	1	1	2.06
s_ν			s_ν^-				1	s_ν	1	2.23
s_ν			s_ν^-				1	1	s_ν	2.06
s_ν			s_ν^-				s_ν	s_ν	1	2.07
s_ν			s_ν^-				s_ν	1	s_ν	2.07
s_ν			s_ν^-				1	s_ν	s_ν	2.07

Table 4.6: Rebound height H_r various magic numbers Λ for St = 152.

Λ	s_v^-	H_r
$3/16$	0.019324	2.07
$1/4$	0.014528	2.01
$1/6$	0.021713	2.04
$1/12$	0.042959	2.07

One parameterization yields an even closer rebound height to the measurements, i.e., $H_r \approx 2.48$, while the others are approximately equal to the TRT-L version.

Furthermore, we tested different values for the magic parameter Λ from Equation (2.12) including the values $1/4$, $1/6$, and $1/12$. The results for the TRT collision operator are shown in Table 4.6. Setting Λ to a different value and such modifying s_v^- has only a minor influence on the rebound height. The same could be observed for other parameterizations apart from the pure TRT.

Additionally, we analyzed many more relaxation rate combinations but could not find a parameter set yielding better results than the one shown in Table 4.5.

4.3.3 Optimizing TRT-C

We showed that choosing the proper relaxation rates is essential to simulate the correct collision behavior. Hence, we also tried unsuccessfully to find a parameter set that might simulate the collision behavior for the TRT-C moments even closer to the measurements from Gondret *et al.* and thus also closer to reality. The only variants

with a notable increase in the rebound height used the bulk viscosity adaption that results in a drastic decrease in numerical stability for high Reynolds numbers.

4.3.4 Influence of Compressibility concerning Sphere-Wall Collision Behavior

Both the cumulants and the MRT-P collision operator are compressible collision operators. As reported above, the MRT-P collision operator is not able to simulate the given sphere-wall collision test case. Apart from different moments and relaxation rates of TRT-C and MRT-P, there is an additional distinction: the MRT-P is a compressible method, while the TRT-C is incompressible. Therefore, it is a reasonable assumption that the difference in compressibility might be the reason for the arising instabilities.

In order to test this, we simulate the given setup for a Stokes number of $\mathrm{St} = 152$ and a sphere diameter of $d_p = 20$ cells for the incompressible and compressible TRT-C collision operator. The resulting gap size trajectories are shown in Figure 4.12, which are identical for both variants. Therefore we do not have to look at any other data but can directly conclude that compressibility at the moment-based collision operators is not influencing the collision behavior.

Therefore, we set the relaxation rates of the MRT-P collision operator to a standard TRT parameterization. The result was a stable simulation with a rebound height of $H_r = 2.27$. Peng *et al.* also used a D3Q19 stencil for their collision operator, but showed in [53] that it was inferior to their D3Q27 stencil. Therefore we decided against further investigating this collision operator.

4.3.5 Parameter Study of the Collision Behavior and the Influence of the Relaxation Rates for Cumulants

For the parameter study regarding the cumulants, we use the D3Q19 stencil instead of the D3Q27 and a particle diameter of $d_p = 10$ cells. There are multiple reasons behind this decision: First, the computational cost is less expensive, which is an which is advantageous when executing many simulations. Second, although we already showed in Section 3.3.3 that the stencil with fewer velocities is significantly worse stability-wise, the potential is to find the proper relaxation rates to make the D3Q19 stencil work and transfer the gained knowledge to the D3Q27 stencil. Looking at the moments of the cumulants for both stencils, given in Appendix A.2, all moments of the D3Q19 stencil are also part of the D3Q27 stencil. With this, we might transfer the knowledge we will gain from analyzing the D3Q19 to the D3Q27 stencil. Third, using the D3Q19, we have fewer free parameters, which makes the potential relaxation rates we have to analyze less. Fourth, both stencils yield qualitatively the same collision behavior, which can be observed in Figure 4.13.

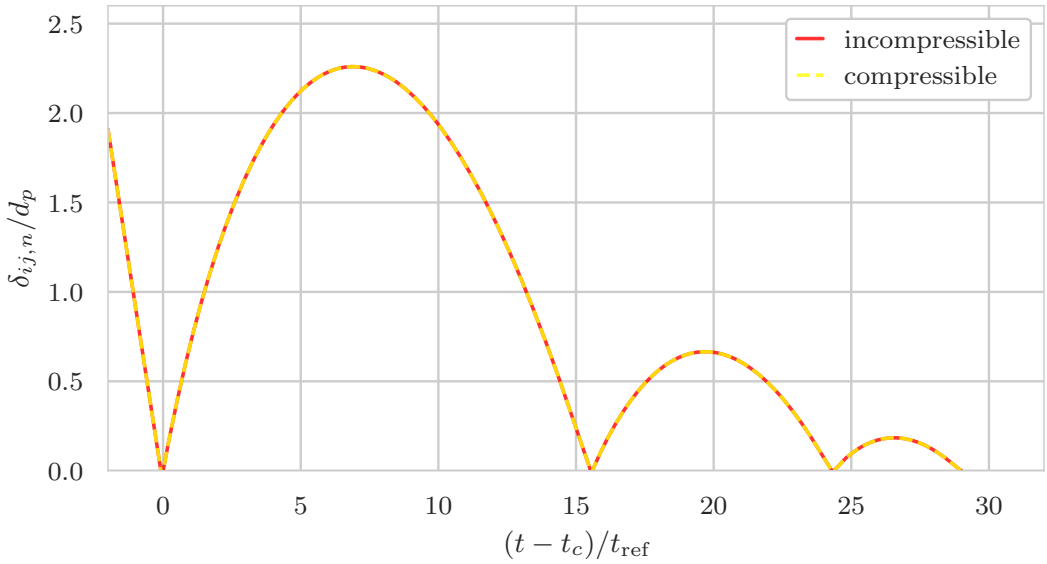
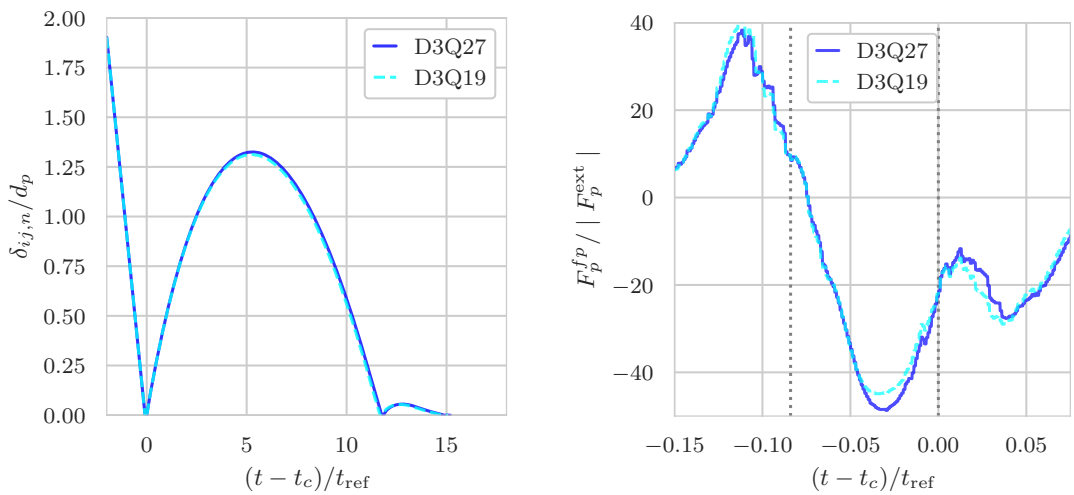


Figure 4.12: Trajectories of normalized particle-wall gap sizes visualized over shifted normalized time for an incompressible and compressible TRT-C collision operator with a Stokes number set to 152.



(a) Gap size trajectories **(b)** Hydrodynamic interaction force

Figure 4.13: Comparison of D3Q19 and D3Q27 stencil regarding the development of gap sizes and hydrodynamic interaction force for St = 152 and $d_p = 20$ cells.

The cumulants implemented in version 0.3.4 of *lbmpy* do not support setting every relaxation rate for every moment separately. Instead, they are grouped, resulting in

$$\{0,\ 0,\ 0,\ 0,\ s_\nu,\ s_\nu,\ s_\nu,\ s_\nu,\ s_\nu,\ \omega_0,\ \omega_1,\ \omega_1,\ \omega_1,\ \omega_2,\ \omega_2,\ \omega_2,\ \omega_3,\ \omega_3,\ \omega_4\}.$$

As for the moment-based collision operators, the first nine relaxation rates are predetermined. Hence, the first four relaxation rates and the following five have to be set to 0 and s_ν respectively to use the proper fluid flow physics for the given setup. This results in ten additional relaxation rates, which are split into five groups, such that ω_x for $x \in [0, 4]$ can be set freely in the range of $(0, 2)$. If we would instead use the D3Q27 stencil, we would have to deal with an additional four relaxation rate groups, and as such, with almost twice the free parameters.

First, we investigate if choosing the collision time $T_c = 2.31\, {}^{d_p}/_{c_s}$ for moment-based collision operators is also the optimal choice regarding the cumulants. There we discovered, that $T_c = 3.46\, {}^{d_p}/_{c_s}$ results in a slightly higher rebound height. For this, we use the same method as Rettinger used in [46]. We run multiple simulations with the same setup, but with a varying collision times. Additionally, we discovered during the early stages of the analysis that disabling the hydrodynamic interaction force during the collision, as used by [9–11], improves the collision behavior from $H_r = 0.95$ to $H_r = 1.19$. This might seem like a small influence, but as we have seen earlier, we had to use multiple small increases to gain a method suitable for the analysis of the collision behavior of spheres.

We started the parameter study for the cumulants, similar as, the moment-based collision operator MRT-L, modifying single relaxation rates and analyzing the resulting rebound heights. One section of this first part can be seen in the left of Table 4.7. Here, setting either $\omega_0 = s_v$ or $\omega_4 = s_v$ makes the simulation unstable, setting $\omega_1 = s_v$ or $\omega_2 = s_v$ produces almost no bounce, and $\omega_3 = s_v$ yields a gap size trajectory closer to the measurements and such yields the only usable parameterization. The reason for the almost non-existent bounce for setting $\omega_1/\omega_2 = s_v$ is again the increase of the hydrodynamic interaction force right before the collision. Even choosing other values, e.g. $\omega_x = y$ for $x \in [0, 1, 2, 4]$, $y \in [0.02, 0.5, 1.5]$ did not change the collision behavior going into the direction of the measurements from Gondret *et al.* The same can be observed for modifying the value for ω_3, compare right part of Table 4.7. As can be seen, increasing or decreasing the value for ω_3 only leads to a damping of the rebound height H_r.

After many more simulations trying different combinations of relaxation rate parameterizations, the only parameter sets improving the collision behavior are the ones shown in Table 4.8. We observe that just setting ω_0 or ω_4 in combination with ω_3 made the simulation unstable, while setting all three increased H_r slightly. Choosing different values for ω_0 and ω_4 leads to the same effect, crashing the simulation, dampening or not changing the rebound height. The only parameter set producing results closer to a more realistic simulation of the collision behavior is the second parameter set of Table 4.8.

Table 4.7: Cumulant relaxation rate parameterization variants and resulting rebound height H_r.

(a) Step 1

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1	1	1	1	1	1.19
s_ν	1	1	1	1	—
1	s_ν	1	1	1	0.072
1	1	s_ν	1	1	0.048
1	1	1	s_ν	1	1.53
1	1	1	1	s_ν	—

(b) Step 2

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1	1	1	1.8	1	1.44
1	1	1	1.85	1	1.47
1	1	1	s_ν	1	1.53
1	1	1	1.99	1	1.53
1	1	1	1.999	1	1.41
1	1	1	1.9999	1	1.18

Table 4.8: Rebound height H_r for different cumulant parameter sets.

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
s_ν	1	1	s_ν	s_ν	1.56
1.999	1	1	s_ν	s_ν	1.67

Table 4.9: Further optimization regarding cumulant relaxation rate parameterization and corresponding rebound height H_r.

(a) Parameter study ω_1

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1.999	0.5	1	s_ν	s_ν	—
1.999	**0.75**	**1**	s_ν	s_ν	**1.78**
1.999	1.25	1	s_ν	s_ν	1.53
1.999	1.5	1	s_ν	s_ν	1.30
1.999	1.75	1	s_ν	s_ν	1.03
1.999	1.9	1	s_ν	s_ν	0.725
1.999	s_ν	1	s_ν	s_ν	0.431
1.999	1.999	1	s_ν	s_ν	0.223

(b) Parameter study ω_2

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1.999	1	0.15	s_ν	s_ν	—
1.999	**1**	**0.25**	s_ν	s_ν	**2.06**
1.999	1	0.5	s_ν	s_ν	1.94
1.999	1	0.75	s_ν	s_ν	1.82
1.999	1	1.25	s_ν	s_ν	1.46
1.999	1	1.5	s_ν	s_ν	1.22
1.999	1	1.75	s_ν	s_ν	—
1.999	1	1.9	s_ν	s_ν	—

At this point, only two relaxations are left to be set: ω_1 and ω_2. Table 4.9 shows the tested parameter sets for ω_1 and ω_2 respectively. Here, the lowest possible value with a stable simulation yields the best fitting rebound height. Increasing the value for either of those two relaxation rates results in a steady damping of H_r until no collision occurs.

Combining these two relaxation rates results in the rebound heights reported in Table 4.10, where we can observe that either choosing $\omega_1 = \omega_2 = 0.5$ or $\omega_1 = 0.25$ and $\omega_2 = 0.75$ produces acceptable results. An unexpected behavior could be observed for the combination shown in the right part of Table 4.10, where setting $\omega_1 = \omega_2 = 1.75$ yields almost no bounce with a rebound height of $H_r = 0.73$ cells, while increasing both relaxation rates at the same time to $\omega_1 = \omega_2 = 1.999$ produces the first rebound height which is higher than the measurements from Gondret *et al.* from all of the tested collision operator parameterizations. An explanation for this behavior is seen in Figure 4.14.

Figure 4.14a visualizes the gap size trajectories from the last reported parameter sets are visualized as well as their gap size trajectories. Standard represents the unmodified cumulants that were tested at the beginning, where all free relaxation rates are set to $\omega_{0-4} = 1$. The other three graphs are named after the setting of ω_1 and ω_2, if only one value is given, both are set to the specific value as reported in Tables 4.11a and 4.11b.

While the parameterization $\omega_1 = \omega_2 = 1.750$ yields almost no bounce, the parameter set $\omega_1 = \omega_2 = 1.999$, which is completely non-intuitive. Following this trend, we expect

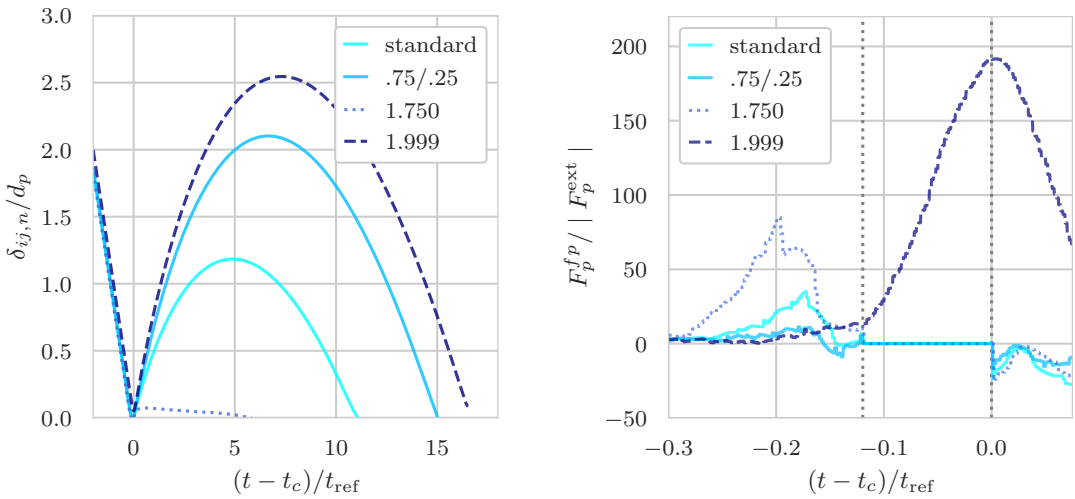
Table 4.10: Best performing relaxation rate parameterization for the cumulants in regard to the rebound height resulting from a sphere-wall collision with a Stokes number of St = 152.

(a) Promising parameter sets

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1.999	0.5	0.5	s_ν	s_ν	2.09
1.999	0.75	0.25	s_ν	s_ν	2.10

(b) Unexpected behavior

ω_0	ω_1	ω_2	ω_3	ω_4	H_r
1.999	1.75	1.75	s_ν	s_ν	0.73
1.999	1.999	1.999	s_ν	s_ν	2.55



(a) Gap size trajectories

(b) Hydrodynamic interaction force

Figure 4.14: Gap size trajectories and hydrodynamic interaction force for the parameter sets given in Tables 4.11a and 4.11b.

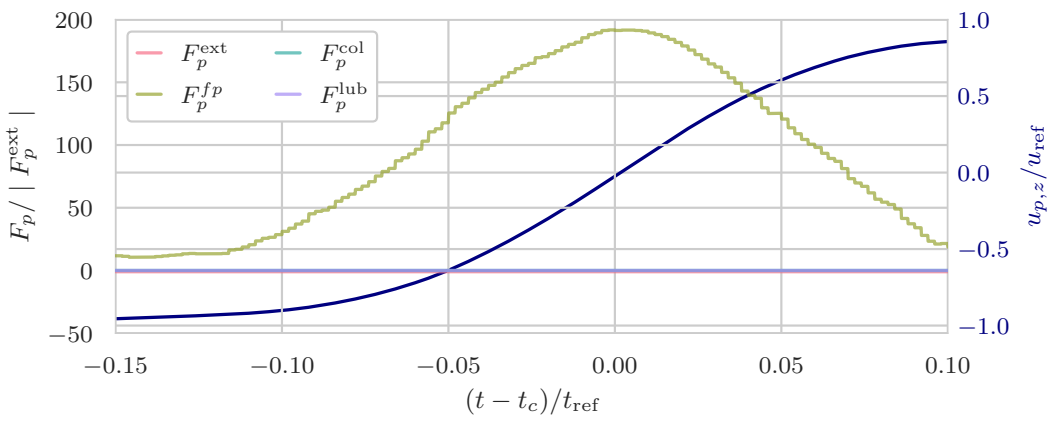


Figure 4.15: Acting forces during "collision" and corresponding particle velocity evolution

the version for 1.999 to have an even smaller bounce than the version for 1.75, but this is not the case. Additionally, observing the trajectory of the 1.999-variant closely, one can see the particle and the wall are never actually in contact, reminding of the collision behavior of the SRT-L. Analyzing the hydrodynamic interaction force reinforces this assumption, because the hydrodynamic interaction should be 0 during a collision event. To be certain, the same process we following analyzing the SRT-L method: we visualize the acting forces during the collision together with the particle velocity evolution, plotted in Figure 4.15. Additionally, we show the lubrication force and collision force, as well as particle velocity in Figure 4.16. As the assumption, that there is no collision event is verified. The difference regarding the rebound height between SRT-L, which is rather small, and the cumulant here is that the collision force is reaching a higher peak and is not dropping during the entire process. Instead it steadily increasing until the moment the particle's velocity reaches zero and afterward declining similarly. Once again the particle comes close to interacting with the wall and activates the lubrication correction model, but moments later is again deactivated before hitting the wall, as seen in Figure 4.16. This behavior is physically not explainable, as no Newtonian fluid can store energy in a way to produce such a "collision" behavior.

As the last step for the cumulant analysis of the sphere wall collision test case with a Stokes number of St = 152, we will investigate if the gained knowledge for the D3Q19 stencil can be transferred to the D3Q27. The moments and corresponding relaxation rates for both stencils are given in Appendix A.2. As described at the beginning of this analysis regarding the cumulants, the moments and corresponding relaxation rates of the D3Q19 stencil are completely contained in the D3Q27 stencil. The D3Q27 cumulant stencil uses four additional relaxation rates that can be set freely in the range of [0,. . . ,2], yielding this five free parameters for the D3Q19 stencil and nine for the D3Q27. These are represented in Table A.1 and Table A.2 for the D3Q19 and D3Q27 stencil, respectively. These free parameters are

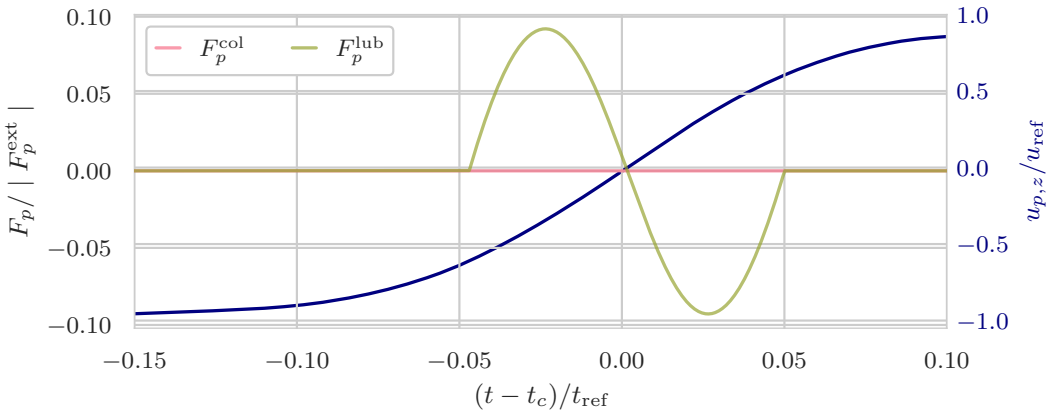


Figure 4.16: Lubrication correction force and collision force during "collision" and resulting velocity development for the particle

- D3Q19 $\Rightarrow \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4\}$
- D3Q27 $\Rightarrow \{\omega_0, \omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6, \omega_7, \omega_8\}$

Here the following applies

$$\omega_{0-2}^{\text{D3Q19}} = \omega_{0-2}^{\text{D3Q27}}, \qquad \omega_{3-4}^{\text{D3Q19}} = \omega_{4-5}^{\text{D3Q27}}.$$

Consequently, $\omega_{3,6-8}$ have to be set in another way. Obeying the time constraints of this thesis, we set $\omega_{3,6-8} = 1$.

For this analysis, we use a diameter of $d_p = 20$ and compare the results for D3Q19 and D3Q27 stencils directly to the measurements of Gondret *et al.*, as illustrated in Figure 4.17. It can be observed that the D3Q19 stencil yields a gap size trajectory, which is much closer to the measurements than the D3Q27 stencil. The rebound trajectory of the D3Q19 stencil matches the measurement data very well. The rebound height is not identical but the collision duration almost is.

4.3.6 Summary and direct Comparison of Results regarding Stokes Number St $= 152$

We will summarize the results and compare them directly against each other. Figure 4.18 shows the gap size trajectories of the various collision operators with the best performing results. All analyzed collision operators were improved due to different modifications to each of them compared to the unmodified parameter sets. Additionally, all TRT parameterizations, as well as the MRT-Lm (modified MRT-L parameter set where $\omega_{0-5} = s_v^-$ $\omega_6 = s_v$), produce almost identical results. Furthermore, although the D3Q19 cumulant collision operator is closest to the measurement data, it is expected that the same can be observed for the TRT-C, MRT-Lm, and TRT-L if the hydrodynamic interaction force is disabled during the collision.

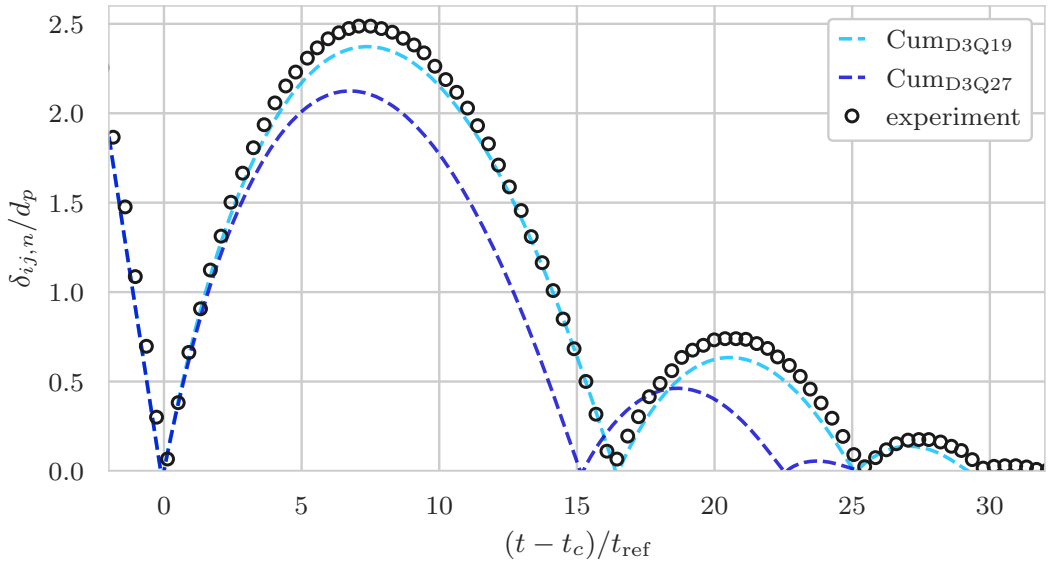


Figure 4.17: Comparison of rebound trajectories of D3Q19 and D3Q27 cumulant collision operator and measured data for a normal collision between sphere and wall with a Stokes number of St = 152.

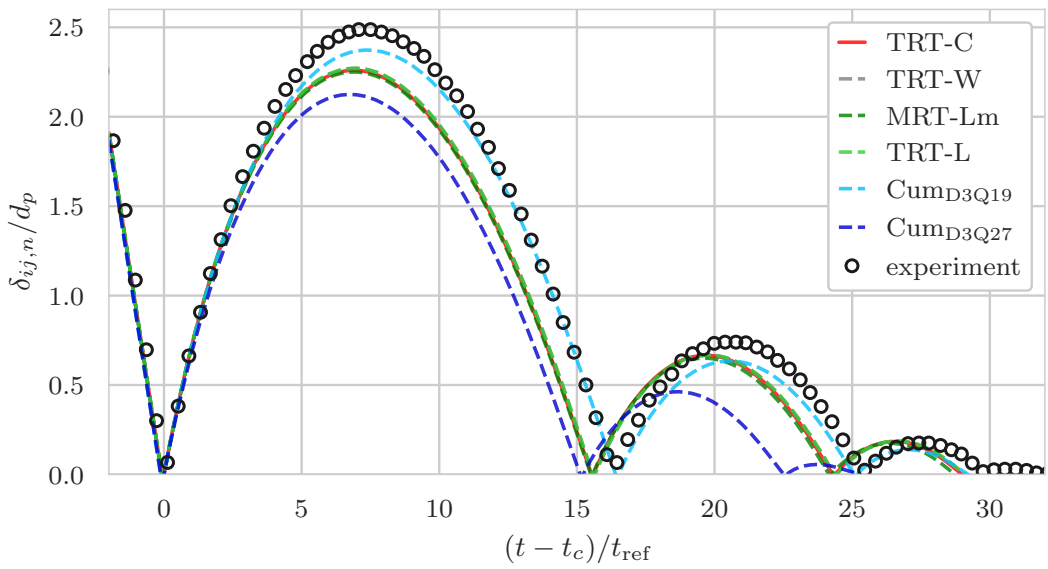


Figure 4.18: Direct comparison of gap size trajectories of all best performing derived collision operators for a sphere-wall collision test with a Stokes number of St = 152.

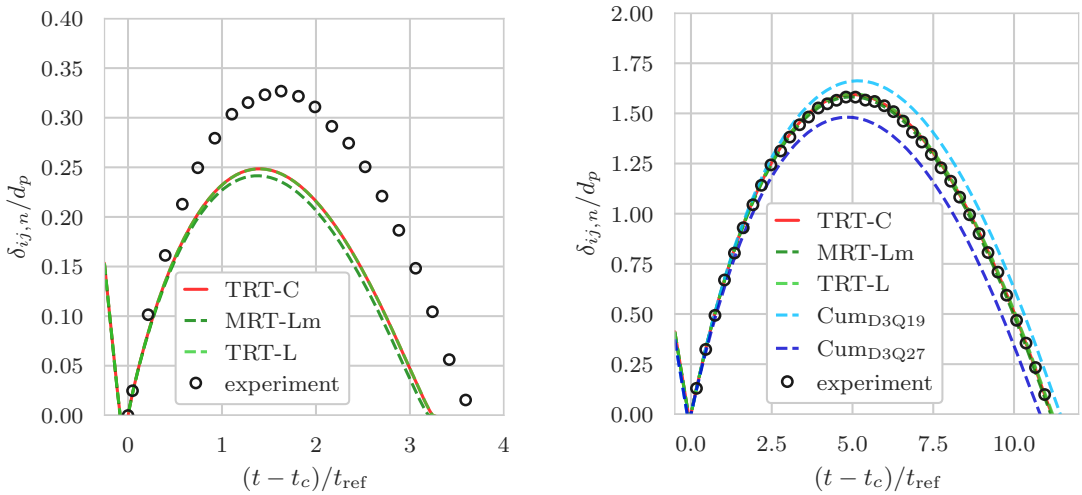
4.3.7 Analyzing St $= 27$ and St $= 100$

We will finish this chapter by briefly analyzing the resulting gap size trajectories for the other two simulation setups given in Table 4.1. Here, we use the same collision operators as used for the comparison regarding a Stokes number of St = 152 directly above. The results are visualized in Figure 4.19a for St = 27 and Figure 4.19b for St = 100.

St $= 27$ Striking here is that only three of the previous six collision operators are present. The reason for the missing of the TRT-P, cumulant D3Q19, and D3Q27 collision operators is that these were unsuccessful in simulating this test case with the given setup, i.e., forces got NaN. Additionally, it can be seen that, again, both TRT parameter sets produce identical gap size trajectories, while the modified MRT-L variant is slightly farther away from the measurement data. Nevertheless, all three variants are not close to the expected trajectories. This is acceptable because we are mainly interested in simulating particulate fluid flow problems with high Reynolds numbers.

Comparing these results with the results produced by Rettinger in [19], it can be assumed that especially for lower Reynolds numbers, the modification of the relaxation rate controlling the bulk viscosity has a significant impact on the collision behavior. This was not further analyzed in the context of this work.

St $= 100$ As Figure 4.19b shows, almost all collision operators could to simulate the given setup with a Stokes number of St = 100 except the TRT-P variant. Here we can observe that the TRT-C, TRT-L, and MRT-Lm are again almost identical. This time, these are almost identical to the measurements from Gondret *et al.* Furthermore, the D3Q19 cumulants slightly over-predict the rebound height, while the D3Q27 slightly under-predict the bounce.



(a) St = 27 **(b)** St = 100

Figure 4.19: Gap size trajectories of the given Stokes numbers and different collision operators.

Chapter 5

Revisiting Numerical Stability Analysis

In this chapter, we analyze the collision operators that could capture the collision behavior of a sphere-wall collision setup with moderate Stokes numbers. To compare their numerical stability, we use the same setup and drastically increase the Stokes number.

5.1 Sphere-Wall Collision with St $= 742$

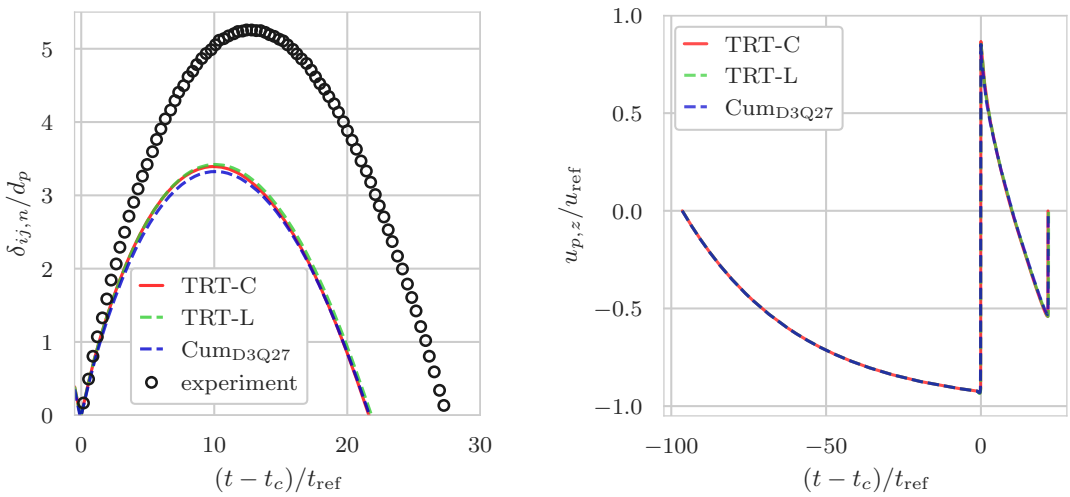
Now that that we know that the TRT-C, TRT-W, MRT-Lm, TRT-L, cumulant D3Q19, and D3Q27 collision operators can simulate the sphere-wall collision test case with sufficient accuracy, at least for Stokes numbers in the range of $100 - 150$, we will try to find their limitations. For this analysis, we again use the sphere-wall collision setup. The Motivation for this decision is, that this test case involves all parts of the four-way coupled algorithm, and can thus be used optimally for numerical stability analysis. Additionally, Gondret *et al.* reported in [56] another parameterization with a much higher Stokes number of St $= 742$. Table 5.1 provides the modified parameters, as well as the used sphere diameter d_p in cells Table 5.1.

5.1.1 Results of the First Attempt

In Figure 5.1, the resulting gap size trajectories and evolution of the particle's velocity are visualized for the collision operators, that successfully finished the simulation. All the missing collision operators, TRT-W, MRT-Lm, and cumulant D3Q19, crashed in

Table 5.1: Sphere-wall-collision parameters used in the experiment by Gondret *et al.* in [56].

St	Re	ρ_f [kg/m³]	μ_f [Ns/m²]	d_p [m]	d_p [cells]	ρ_p [kg/m³]	domain size [d_p]
742	788	920	$5e^{-3}$	0.005	20	7800	$12 \times 12 \times 80$



(a) Gap size trajectories **(b)** Particle velocity evolution

Figure 5.1: Simulation data of an attempted sphere-wall collision setup with a Stokes number of $St = 742$.

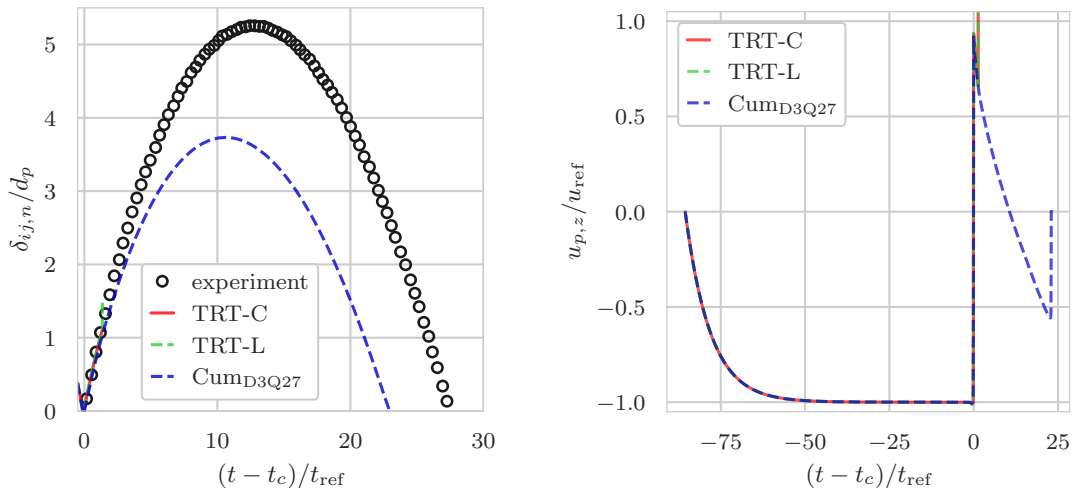
the first $2.5e^4$ time steps due to NaN values detected in the part of the algorithm responsible for the rotation[1]. As a reference, the simulations that finished successfully took about $1.2e^5$ time steps.

Now, let us look at the simulations results reported in Figure 5.1. As shown in Figure 5.1a, the gap size trajectories of the functioning collision operators are not close to the measurements from Gondret *et al.* This can be explained by looking at the development of the particle's velocity visualized in Figure 5.1b. Here, we can observe, that the expected terminal velocity of 0.02 lattice units is not reached. The maximal recorded particle velocity for all variants was ≈ 0.0187 lattice units. Taking this velocity to calculate the Stokes number and Reynolds number to compare this simulation to other, results in St ≈ 690, and Re ≈ 730. To fix this problem, we have two possibilities:

- expand the simulation domain in the z-direction, or
- modifying the artificial acceleration of the sphere.

In order to not increase the computational cost further, we chose the second option, and increased the acceleration by a factor of 5.

[1]This could not be further investigated.



(a) Gap size trajectories **(b)** Particle velocity evolution

Figure 5.2: Simulation data of a sphere-wall collision setup with a Stokes number of $St = 742$.

5.1.2 Results of the Second Attempt

The simulation results for the gap size trajectories and developing particle velocity are given in Figure 5.2. As depicted in Figure 5.2b, we reach the terminal velocity of 0.02 lattice units for all variants. Although only the cumulant D3Q27 collision operator could finish this simulation successfully, but still yields a gap size trajectory far from the measured data. We will investigate this behavior further, after analyzing the two crashed versions.

Both TRT-C and TRT-L[2] crash shortly after the collision, because the velocity explodes. Consequently, we can conclude, that the five-times higher sphere acceleration cannot cause this behavior. Investigating this further by looking at the acting forces, plotted in Figure 5.3, we see that forces are not responsible for the exploding velocity. Here, the reason for the simulation crash, is again reported values in the rotation part of the algorithm.

At last, we take a closer look at the forces and resulting particle velocity, shown in Figure 5.4 during the time period around the collision, to gain further insights, as why the collision behavior is again not meeting expectations. Again we observe that the hydrodynamic interaction force increases shortly before the collision. Hence, the particle's velocity drops slightly, such that the collision velocity is not as high as expected. The same behavior was reported for the unmodified cumulants. The

[2]Here, we see the results for the TRT-L D3Q27 collision operator. The TRT-L with 19 discrete velocities performed almost identical to the TRT-C D3Q19.

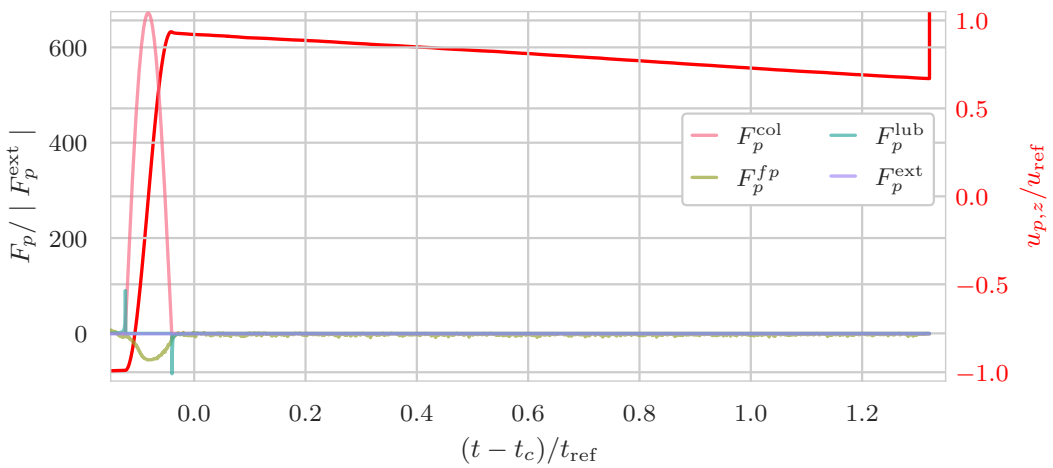


Figure 5.3: Acting forces and corresponding development of particle's velocity for the TRT-C collision operator and a Stokes number of $St = 742$. Time period is shortly before the collision until the crash of the simulation.

introduced modifications work for smaller Stokes numbers, but have to be further investigated for larger ones, as these results show.

5.2 Further Studies with St $= 942$

Here, we used the same parameters as reported in Table 5.1 and increased the Stokes number to St $= 942$. Unfortunately, we used the incorrect acceleration factor, such that the terminal velocity of 0.02 lattice units could not be reached[3], as it can be seen in Figure 5.5b. Evaluating the particle's velocity before the collision results in a Stokes number of St $= 869$ and Reynolds number of Re $= 923$. The resulting gap size trajectories are visualized in Figure 5.5a. Additionally, we plotted the acting forces around the collision in Figure 5.6, where we can again observe the increase of the hydrodynamic interaction force before the collision. With this, we gathered three simulation data to compare the development of the hydrodynamic force in the next section.

5.3 Comparison of Maximal Hydrodynamic Interaction Force for Different Stokes Numbers

In Table 5.2, the maximal values for the normalized hydrodynamic interaction force $F_{p,\max}^{fp}$ in lattice units ahead of the collision are reported for the cumulant D3Q27 and

[3]We were not able to re-run this simulation due to time constraints.

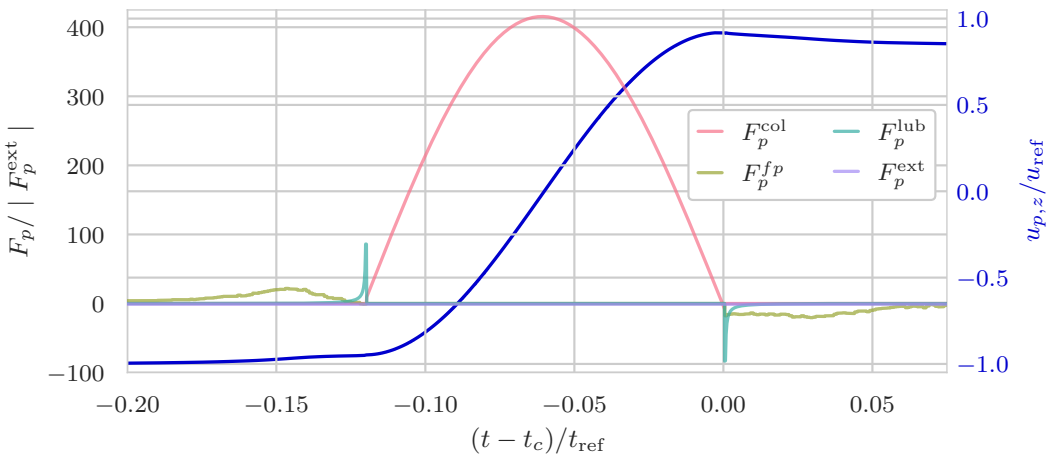
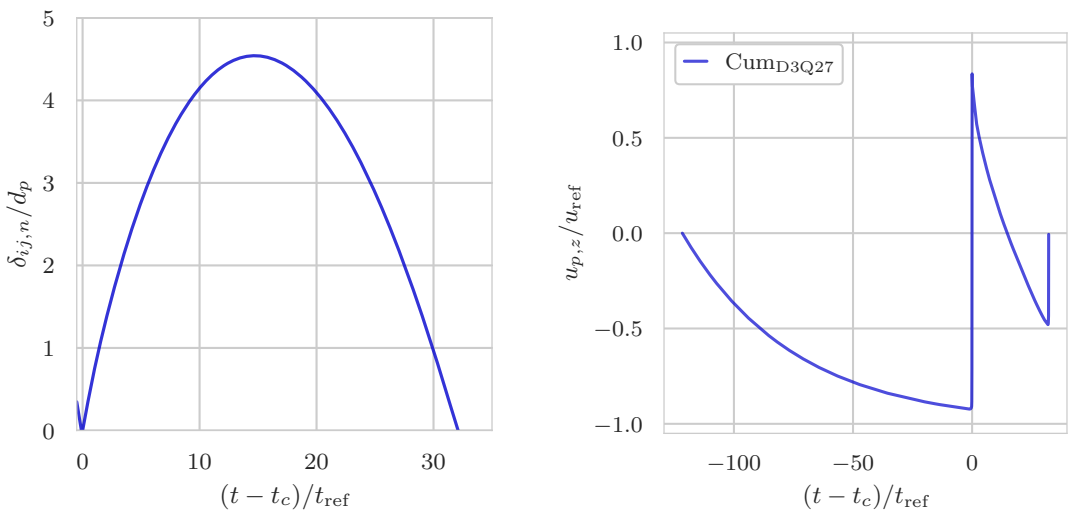


Figure 5.4: Acting forces and corresponding development of particle's velocity for the cumulant D3Q27 collision operator and a Stokes number of $St = 742$. Time period plotted is before, during and shortly after the collision event.



(a) Gap size trajectories

(b) Particle velocity evolution

Figure 5.5: Simulation data of a sphere-wall collision setup with a Stokes number of $St = 942$.

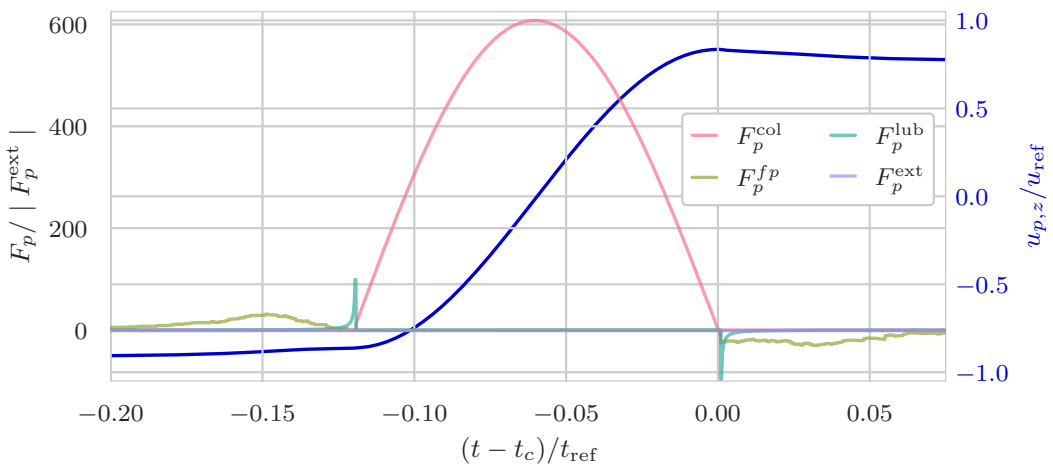


Figure 5.6: Acting forces and corresponding development of particle's velocity for the TRT-C collision operator and a Stokes number of $St = 942$ for a duration around the collision event.

Table 5.2: Maximal values for normalized hydrodynamic interaction force ahead of the collision for previous test cases with high Stokes and Reynolds numbers.

St	Re	$F_{p,\max}^{fp,\text{cumulant}}/F_p^{\text{ext}}$	$F_{p,\max}^{fp,\text{TRT-C}}/F_p^{\text{ext}}$
690	730	20.01	9.06
742	788	21.84	9.57
892	923	31.23	—

TRT-C collision operators. We observe that the maximal values for the normalized hydrodynamic interaction force are higher in all cases than the TRT-C forces. Also, it seems that the normalized forces increase exponentially with higher Stokes numbers. Hence, we assume the collision behavior is deviating further from reality with increasing Stokes numbers. Consequently, we can conclude that the cumulant D3Q27 collision operator is stable even for high Stokes and Reynolds numbers. However, by using them, one must be aware of unsatisfying capturing regarding the collision behavior for high Stokes and Reynolds numbers.

Chapter 6

Conclusion and Outlook

In this work, we first analyzed the four-way coupled algorithm developed by Rettinger regarding its numerical stability for high Reynolds numbers. Earlier studies showed, that the algorithm with the TRT+aB collision operator of the LBM has limitations concerning numerical stability for particle-laden flow simulations with high Reynolds numbers. Therefore, we studied the current state and identified that the collision operator plays a vital role in the numerical stability of such simulations. Consequently, we tested multiple collision operators with a standard parameterization of the relaxation rates in a sphere-wall collision setup with moderate Stokes numbers. Here, we found that the values of the relaxation rates of the higher-order moments have a large influence on the collision behavior. Setting them incorrectly can lead to completely non-physical behavior, where, e.g., sphere and wall are never in contact. Due to a rapid increase of the hydrodynamic interaction force, the sphere is stopped and reflected before contact. Consequently, we performed multiple time-consuming and thorough parameter studies to find a proper parameter set for the relaxation rates of the used collision operators to enable an acceptable accuracy of the collision behavior.

After modifying the collision operators and their relaxation rates, they were compared in a test case with high Stokes numbers. There, the cumulants produced promising results regarding their numerical stability. Nevertheless, especially, the setting of the relaxation rates of the D3Q27 stencil requires additional testing and modification to be usable for particulate flows with high Stokes numbers. Without further modification, this collision operator is barely usable for such simulations because of its non-physical representation of the hydrodynamic interaction force shortly before the collision. We observed the same behavior for the standard parameterization of the cumulants for moderate Stokes numbers. Therefore, we assume it is possible to further increase its accuracy regarding the hydrodynamic interaction force. This process showed how essential a thorough analysis and comparison to real-world measurements are.

We identified multiple prospects for future work that could be further analyzed in this thesis. In this thesis, we identified multiple prospects that remain to be further analyzed:

Central Moments Central moments are an advanced and promising collision operator that was added in version 4.0 to *lbmpy*. Central moments were developed by Geier *et al.* in [57] and are said to be comparably stable as the cumulants.

Instabilities Rotation The analysis in Chapter 5 showed numerical instabilities arising multiple times in the part of the coupled-algorithm responsible for the handling of the rotation. These numerical instabilities, occurred, especially, at the beginning of the simulation or shortly after a collision event.

Cumulants In order to use them in particulate flow simulations with high Reynolds numbers, an additional modification regarding the relaxation rates of the D3Q27 stencil is necessary.

Many Particles In this work, we could unfortunately not test the collision operators in environments with many interacting particles. Such a setup would have been disadvantageous for the validation and calibration in this work. Nevertheless, the algorithm with the modified collision operator has to be tested in actual particulate flows, where numerous particles are interacting.

Appendix

A Moments of the LBM Models

General notes regarding moment transformation matrices The moment transformation matrix appears in Equation (2.7) and transforms PDFs into the moment space for moment based collision operators. The ordering of the lattice velocities c_q are uniquely defined by the second, third and fourth row of the corresponding moment transformation matrix.

A.1 TRT-C

The moment transformation matrix $\boldsymbol{M}$ for the TRT-C collision operator (on the basis of the MRT model derived in [22]) is given as

$$\boldsymbol{M} = \begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 \\
0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
1 & -2 & -2 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & 2 & -2 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 \\
0 & -2 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -2 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
0 & -1 & -1 & 2 & 2 & -1 & -1 & 1 & 1 & 1 & 1 & -2 & -2 & 1 & 1 & -2 & -2 & 1 & 1 \\
0 & 1 & 1 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & -1 \\
0 & 1 & 1 & -2 & -2 & 1 & 1 & 1 & 1 & 1 & 1 & -2 & -2 & 1 & 1 & -2 & -2 & 1 & 1 \\
0 & -1 & -1 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1
\end{bmatrix}$$

with the corresponding values for the equilibrium moments

$$\boldsymbol{m}^{\mathrm{eq}} = \left(\rho, u_x, u_y, u_z, u_x^2 + u_y^2 + u_z^2, 0, 0, 0, 0, 2u_x^2 - u_y^2 - u_z^2, u_y^2 - u_z^2, u_x u_y, u_y u_z, u_x u_z, 0, 0, 0, 0, 0\right)^{\mathrm{T}}.$$

A.2 Cumulants

Due to the non-linear transformation of the central moments to cumulants we cannot give a moment transformation matrix, instead we provide the cumulants with corresponding equilibrium values and relaxation rates. These are given for the D3Q19 cumulant stencil in Table A.1 and for the D3Q27 in Table A.2.

With the code given in Listing A.1 the cumulant LBM kernel is generated using *lbmpy* version 0.3.4. We directly set the relaxation rates depending on the viscosity of the fluid accordingly to s_v, to directly indicate these cannot be freely set.

```
import sympy as sp

from lbmpy.session import *

from pystencils_walberla import CodeGeneration
from pystencils_walberla import get_vectorize_instruction_set

with CodeGeneration() as ctx:
  cpu_vectorize = {'instruction_set': get_vectorize_instruction_set(ctx)}

  omega = sp.symbols('omega_:10')  # for the D3Q19 stencil use 'omega_:6'
  stencil = get_stencil('D3Q27', 'walberla')  # or 'D3Q19'
  layout = 'fzyx'

  optimizations = {'cse_global': True}
  coll_params = {'stencil': stencil,
                 'method': 'cumulant',
                 'relaxation_rates': omega,
                 'galilean_correction': True if stencil == 'D3Q27'
                 else False}

  collision_rule = create_lb_collision_rule(optimization=optimizations,
                                            **method_params)

  generate_lattice_model(CodeGeneration(), 'GeneratedLBM', collision_rule,
                         field_layout=layout,
                         cpu_vectorize_info=cpu_vectorize)
```

Listing A.1: Generation code of cumulant LBM kernel

Table A.1: *lbmpy* method of D3Q19 cumulants

Central Moment / Cumulant	Equilibrium Value	Relaxation Rate
1 (central moment)	ρ	0
x (central moment)	0	0
y (central moment)	0	0
z (central moment)	0	0
xy	0	s_v
xz	0	s_v
yz	0	s_v
$x^2 - y^2$	0	s_v
$x^2 - z^2$	0	s_v
$x^2 + y^2 + z^2$	ρ	ω_0
$xy^2 + xz^2$	0	ω_1
$x^2y + yz^2$	0	ω_1
$x^2z + y^2z$	0	ω_1
$xy^2 - xz^2$	0	ω_2
$x^2y - yz^2$	0	ω_2
$x^2z - y^2z$	0	ω_2
$x^2y^2 - 2x^2z^2 + y^2z^2$	0	ω_3
$x^2y^2 + x^2z^2 - 2y^2z^2$	0	ω_3
$x^2y^2 + x^2z^2 + y^2z^2$	0	ω_4

Table A.2: *lbmpy* method of D3Q27 cumulants

Central Moment / Cumulant	Equilibrium Value	Relaxation Rate
1 (central moment)	ρ	0
x (central moment)	0	0
y (central moment)	0	0
z (central moment)	0	0
xy	0	s_v
xz	0	s_v
yz	0	s_v
$x^2 - y^2$	0	s_v
$x^2 - z^2$	0	s_v
$x^2 + y^2 + z^2$	ρ	ω_0
$xy^2 + xz^2$	0	ω_1
$x^2y + yz^2$	0	ω_1
$x^2z + y^2z$	0	ω_1
$xy^2 - xz^2$	0	ω_2
$x^2y - yz^2$	0	ω_2
$x^2z - y^2z$	0	ω_2
xyz	0	ω_3
$x^2y^2 - 2x^2z^2 + y^2z^2$	0	ω_4
$x^2y^2 + x^2z^2 - 2y^2z^2$	0	ω_4
$x^2y^2 + x^2z^2 + y^2z^2$	0	ω_5
x^2yz	0	ω_6
xy^2z	0	ω_6
xyz^2	0	ω_6
x^2y^2z	0	ω_7
x^2yz^2	0	ω_7
xy^2z^2	0	ω_7
$x^2y^2z^2$	0	ω_8

A.3 MRT-L

A.3.1 D3Q19 Stencil

The moment transformation matrix $\boldsymbol{M}$ for the D3Q19 MRT-L and TRT-L collision operators is given as

$$\boldsymbol{M} = \begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 \\
0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
0 & -1 & -1 & 2 & 2 & -1 & -1 & 1 & 1 & 1 & 1 & -2 & -2 & 1 & 1 & -2 & -2 & 1 & 1 \\
0 & 1 & 1 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & -1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & -1 & 1 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 2 & 2 & 1 & 1 & -2 & -2 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & -2 & 2 & -2 & 2 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 \\
0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 \\
0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 \\
1 & -5 & -5 & -5 & -5 & 4 & 4 & 7 & 7 & 7 & 7 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 \\
1 & 2 & 2 & -5 & -5 & -3 & -3 & 0 & 0 & 0 & 0 & -2 & -2 & 5 & 5 & -2 & -2 & 5 & 5 \\
1 & -3 & -3 & 0 & 0 & -3 & -3 & 0 & 0 & 0 & 0 & 3 & 3 & 0 & 0 & 3 & 3 & 0 & 0
\end{bmatrix}$$

and its corresponding equilibrium values are

$$\boldsymbol{m}^{\mathrm{eq}} = \left(\rho, u_x, u_y, u_z, 2u_x^2 - u_y^2 - u_z^2, u_x^2 - u_y^2, u_x u_y, u_x u_z, u_y u_z, u_x^2 + u_y^2 + u_z^2, 0, 0, 0, 0, 0, 0, 0, 0, 0\right)^{\mathrm{T}}$$

A.3.2 D3Q27 Stencil

The moment transformation matrix $\boldsymbol{M}$ for the D3Q27 MRT-L and TRT-L collision operators is given as

$$\boldsymbol{M} = \begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
0 & -1 & -1 & 2 & 2 & -1 & -1 & 1 & 1 & 1 & 1 & -2 & -2 & 1 & 1 & -2 & -2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 1 & 0 & 0 & -1 & -1 & 1 & 1 & 1 & 1 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\
0 & -1 & 1 & 0 & 0 & 0 & 0 & 2 & 2 & -2 & -2 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & 2 & 2 & -2 & -2 & 2 & 2 & -2 & -2 \\
0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 2 & 2 & 1 & 1 & -2 & -2 & 2 & 2 & 2 & 2 & -2 & -2 & -2 & -2 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & -2 & 2 & -2 & 2 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 2 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & -2 & 2 & 0 & 0 & -2 & 2 & 2 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\
0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 2 & 2 & -1 & -1 & -2 & -2 & 1 & 1 & 2 & 2 & 2 & 2 & -2 & -2 & -2 & -2 \\
0 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 2 & -2 & 0 & 0 & 2 & -2 & 0 & 0 & 2 & 2 & -2 & -2 & 2 & 2 & -2 & -2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\
1 & -2 & -2 & -2 & -2 & 1 & 1 & 4 & 4 & 4 & 4 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & 4 & 4 & 4 & 4 & 4 & 4 & 4 & 4 \\
1 & 1 & 1 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & 4 & 4 & -2 & -2 & 4 & 4 & 4 & 4 & 4 & 4 & 4 & 4 & 4 & 4 \\
1 & -2 & -2 & 1 & 1 & -2 & -2 & -2 & -2 & -2 & -2 & 4 & 4 & -2 & -2 & 4 & 4 & -2 & -2 & 4 & 4 & 4 & 4 & 4 & 4 & 4 & 4 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 & -1 & 0 & 0 & 2 & 2 & -2 & -2 & -2 & -2 & 2 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & -1 & 1 & 2 & -2 & 2 & -2 & -2 & 2 & -2 & 2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & -2 & -2 & 2 & 2 & -2 & -2 & 2 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & -2 & -2 & -2 & -2 & 2 & 2 & 2 & 2 & 4 & 4 & 4 & 4 & -4 & -4 & -4 & -4 \\
0 & 1 & -1 & 0 & 0 & 0 & 0 & -2 & -2 & 2 & 2 & -2 & 2 & 0 & 0 & -2 & 2 & 0 & 0 & 4 & 4 & -4 & -4 & 4 & 4 & -4 & -4 \\
0 & 0 & 0 & -1 & 1 & 0 & 0 & 2 & -2 & 2 & -2 & 0 & 0 & 2 & -2 & 0 & 0 & 2 & -2 & 4 & -4 & 4 & -4 & 4 & -4 & 4 & -4 \\
-1 & 2 & 2 & 2 & 2 & 2 & 2 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & -4 & 8 & 8 & 8 & 8 & 8 & 8 & 8 & 8
\end{bmatrix}.$$

Its corresponding equilibrium values are given as

$$\boldsymbol{m}^{\mathrm{eq}} = \left(\rho, u_x, u_y, u_z, 2u_x^2 - u_y^2 - u_z^2, u_x^2 - u_y^2, u_x u_y, u_x u_z, u_y u_z, u_x^2 + u_y^2 + u_z^2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0\right)^{\mathrm{T}}.$$

A.4 MRT-P

The moment transformation matrix $\boldsymbol{M}$ for the D3Q27 MRT-P collision operator derived by Peng in [55] is given as

$$\boldsymbol{M} = \begin{bmatrix}
1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
-2 & -1 & -1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\
0 & 2 & 2 & -1 & -1 & -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -2 & -2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\
0 & -4 & 4 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 2 & 2 & 2 & 2 & -2 & -2 & -2 & -2 \\
0 & 0 & 0 & -4 & 4 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & 0 & 0 & -1 & -1 & 1 & 1 & 2 & 2 & -2 & -2 & 2 & 2 & -2 & -2 \\
0 & 0 & 0 & 0 & 0 & -4 & 4 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & 2 & -2 & 2 & -2 & 2 & -2 & 2 & -2 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \\
8 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\
0 & -4 & -4 & 2 & 2 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & -2 & -2 & -2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -2 & -2 & 2 & 2 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 2 & 2 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 2 & 2 & -2 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2 & 2 & 2 & -2 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\
0 & 4 & -4 & 0 & 0 & 0 & 0 & -2 & -2 & 2 & 2 & -2 & -2 & 2 & 2 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\
0 & 0 & 0 & 4 & -4 & 0 & 0 & -2 & 2 & -2 & 2 & 0 & 0 & 0 & 0 & -2 & -2 & 2 & 2 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\
0 & 0 & 0 & 0 & 0 & 4 & -4 & 0 & 0 & 0 & 0 & -2 & 2 & -2 & 2 & -2 & 2 & -2 & 2 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\
-8 & 4 & 4 & 4 & 4 & 4 & 4 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1
\end{bmatrix}$$

with the corresponding equilibrium moments

$$\begin{aligned}
\boldsymbol{m}^{\mathrm{eq}} = \Big(&\rho, u_x, u_y, u_z, -\rho u_x^2 - u_y^2 - u_z^2, 2u_x^2 - u_y^2 - u_z^2, u_x^2 - u_y^2, u_x u_y, u_y u_z, u_x u_z, -2u_x, -2u_y, -2u_z, 0, 0, 0, 0, 2\rho - 4u_x^2 - 4u_y^2 - 4u_z^2, \\
&- 2u_x^2 + u_y^2 + u_z^2, -u_y^2 + u_z^2, -u_x u_y, -u_y u_z, -u_x u_z, u_x, u_y, u_z, -\rho + 3u_x^2 + 3u_y^2 + 3u_z^2, \Big)^{\mathrm{T}}.
\end{aligned}$$

Appendix

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